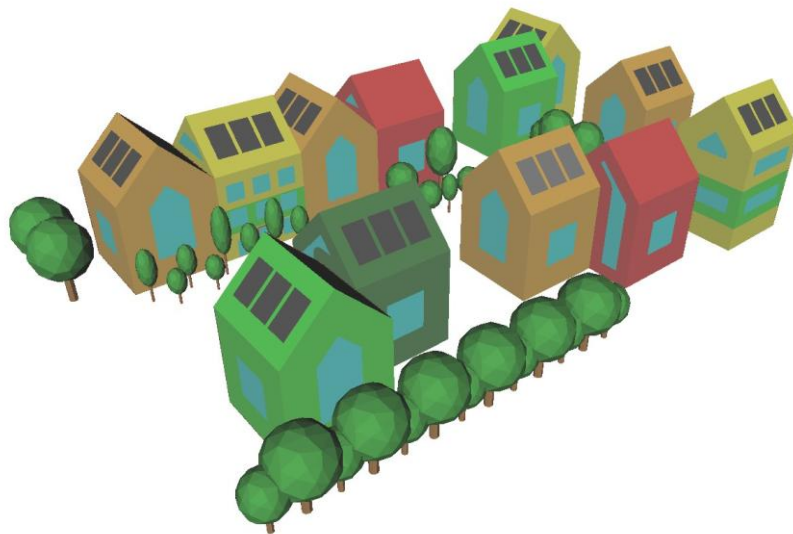




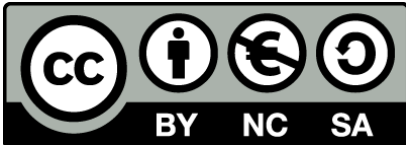
Beta 8

UML diagrams



Last updated on 26 March 2026





This work is licensed under the [Creative Commons License CC BY-NC-SA 4.0](https://creativecommons.org/licenses/by-nc-sa/4.0/) licence.

The latest version of this document can be downloaded from:

https://github.com/tudelft3d/Energy_ADE/tree/3.0beta8/documentation

Author

Giorgio Agugiaro
3D Geoinformation Group
Department of Urbanism
Faculty of Architecture and Built Environment
Delft University of Technology
The Netherlands
g.agugiaro@tudelft.nl
<https://3d.bk.tudelft.nl/gagugiaro/>

Funding

Development of the Energy ADE 3.0 has started in spring 2024 within the European *DigiTwins4PEDs* project (Driving Urban Transition call 2022), co-funded by the European Commission under the Horizon Europe Partnership scheme.

Development is now continuing within the project *RenoDAT*, funded by the Dutch Research Council, and the European project *FlexPED*, (Driving Urban Transition call 2024), co-funded by the European Commission under the Horizon Europe Partnership scheme.

Since 2025, several other partners have joined on a volunteer basis as contributors, early testers or technology enthusiasts, who further develop the Energy ADE as a joint, open project.

Acknowledgements

The author would like to thank the following people for their suggestions and fruitful discussions that have helped design the current version of the Energy ADE conceptual data model:

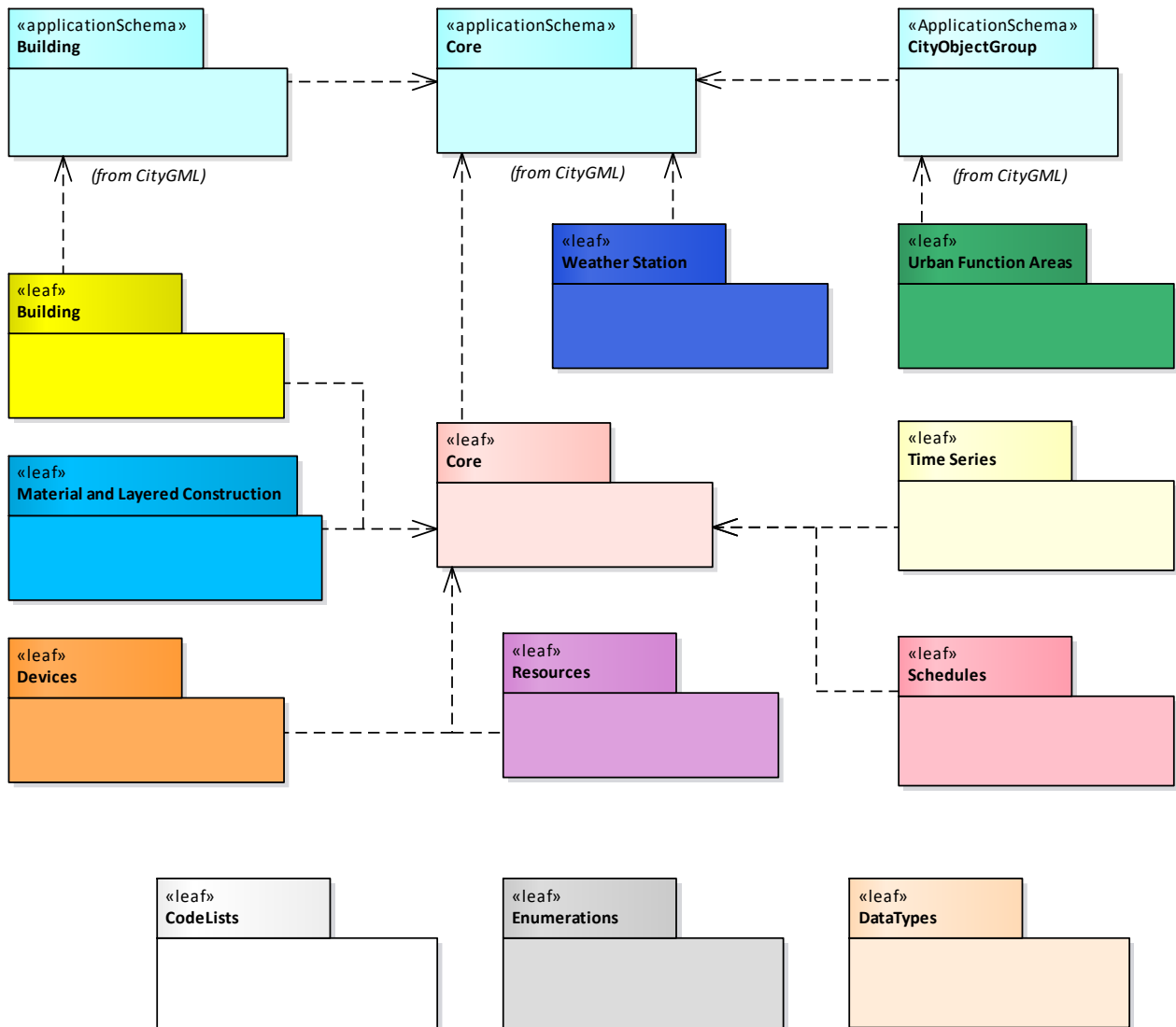
- Rushikesh Padsala, Matthias Betz, Zhihang Yao and Volker Coors (HFT Stuttgart, Germany)
- Jérôme Kämpf (Idiap, Switzerland)
- Thomas Kolbe (TU Munich, Germany)
- Camilo León-Sánchez, Hiba Doi, Daan Schlosser (TU Delft, The Netherlands)
- Maxim Shamovic and Simon Raming (RWHT Aachen, Germany)
- Karl Heinz Häfele and Andreas Geiger (KIT, Germany)

Particular gratitude is due to Daniele Gasparini for designing the new Energy ADE logo.

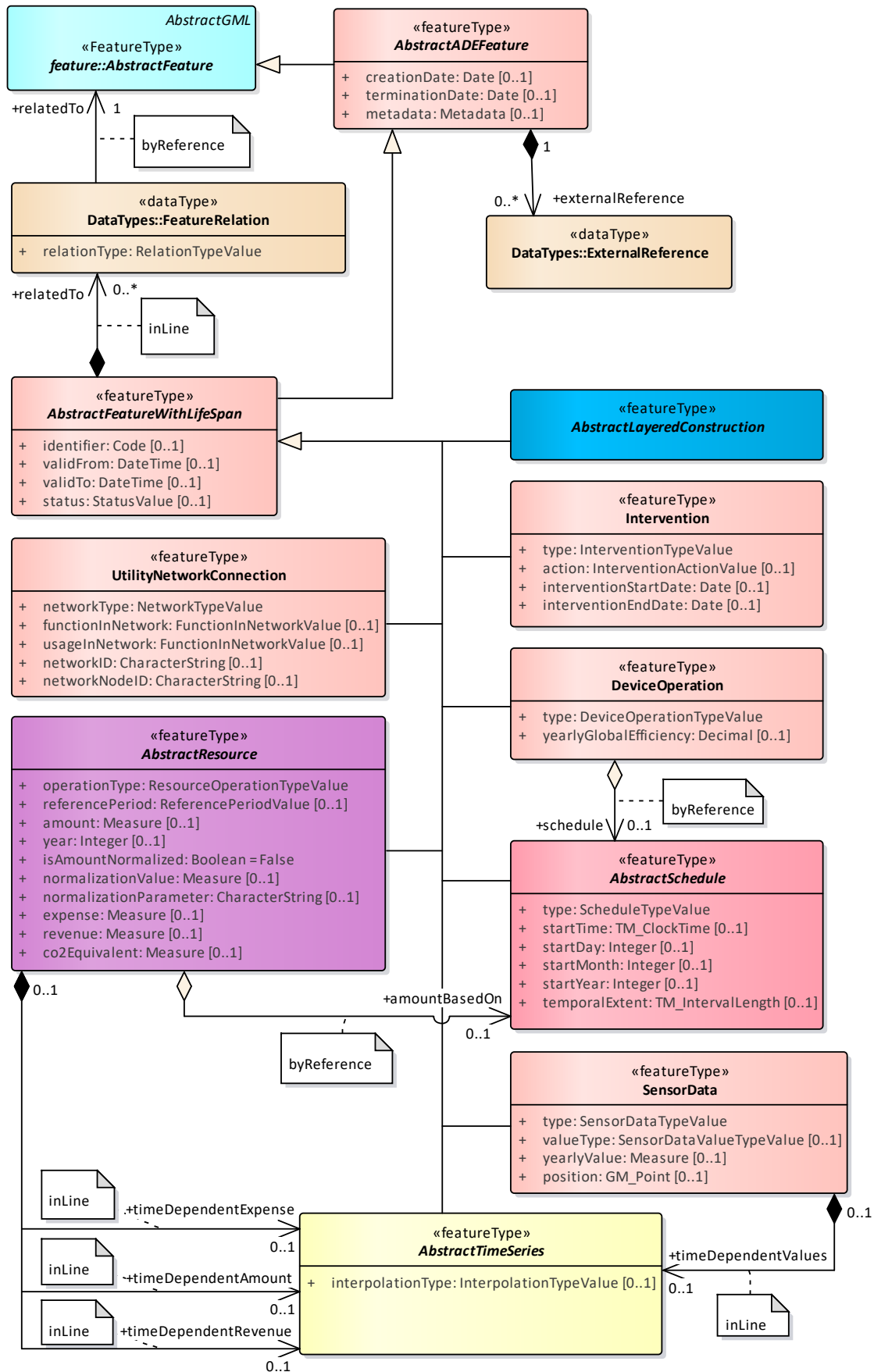
Contents

Overview of the packages.....	3
Core package part 1	4
Core package part 1: CodeLists and Enumerations	5
Core package part 2	6
Core package part 2: CodeLists and Enumerations	7
Building package part 1	8
Building package part 1: CodeLists and Enumerations	9
Building package part 2	10
Building package part 2: CodeLists and Enumerations	11
Building package part 3	12
Building package part 3: CodeLists and Enumerations	13
Devices package part 1	14
Devices package part 1: CodeLists and Enumerations	15
Devices package part 2	16
Devices package part 2: CodeLists and Enumerations	17
Urban function areas package and CodeLists	18
Weather station package and CodeLists	18
Resources package.....	19
Resources package: CodeLists and Enumerations	20
Layered construction package.....	21
Layered construction package: CodeLists	22
Schedules package.....	23
Schedules package: CodeLists	24
Time series package part 1 and CodeLists.....	25
Time series package part 2.....	26
DataTypes package.....	27
DataTypes package: CodeLists and Enumerations	28
Enumerations package	29
Codelists package	30
Changelog	35

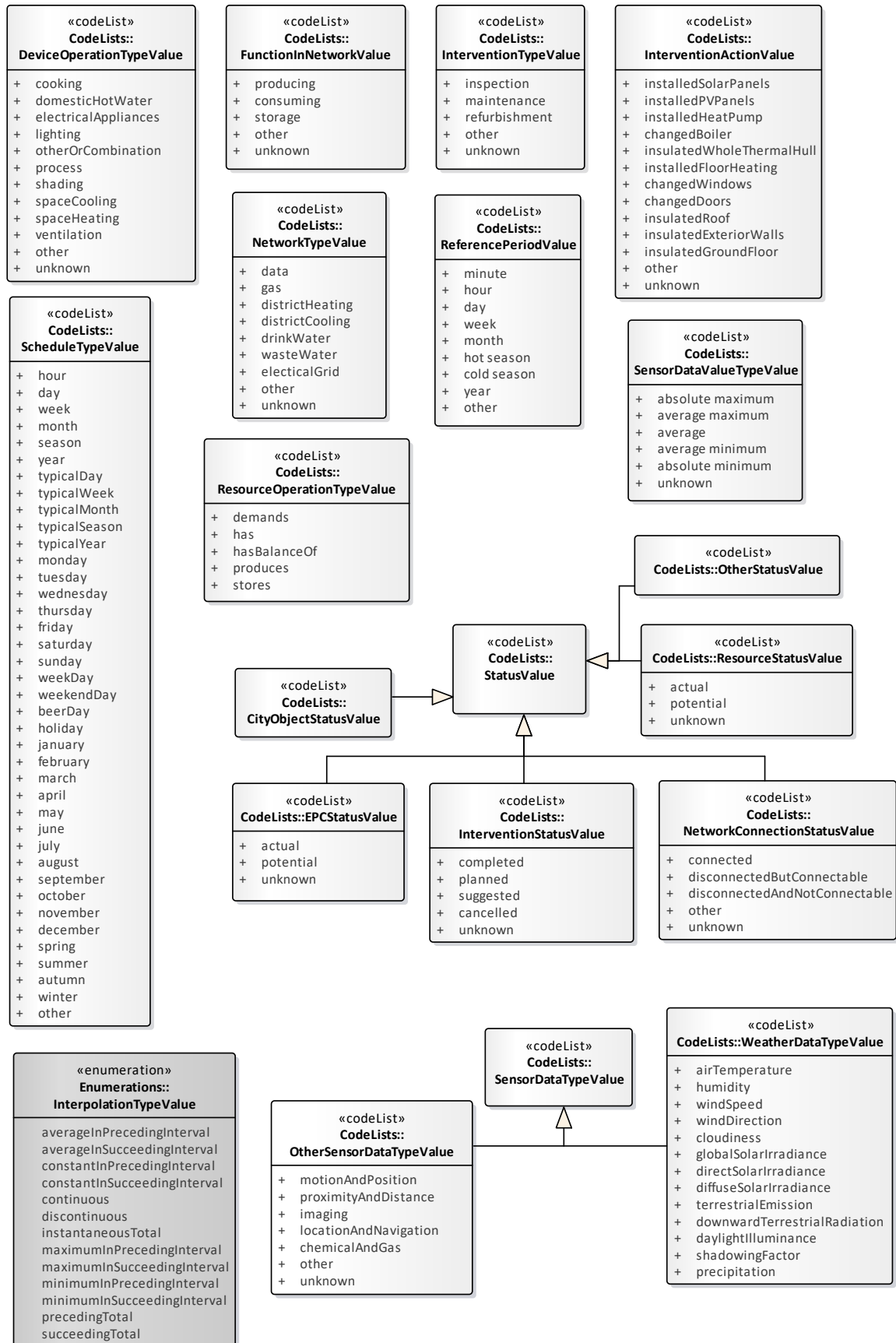
Overview of the packages



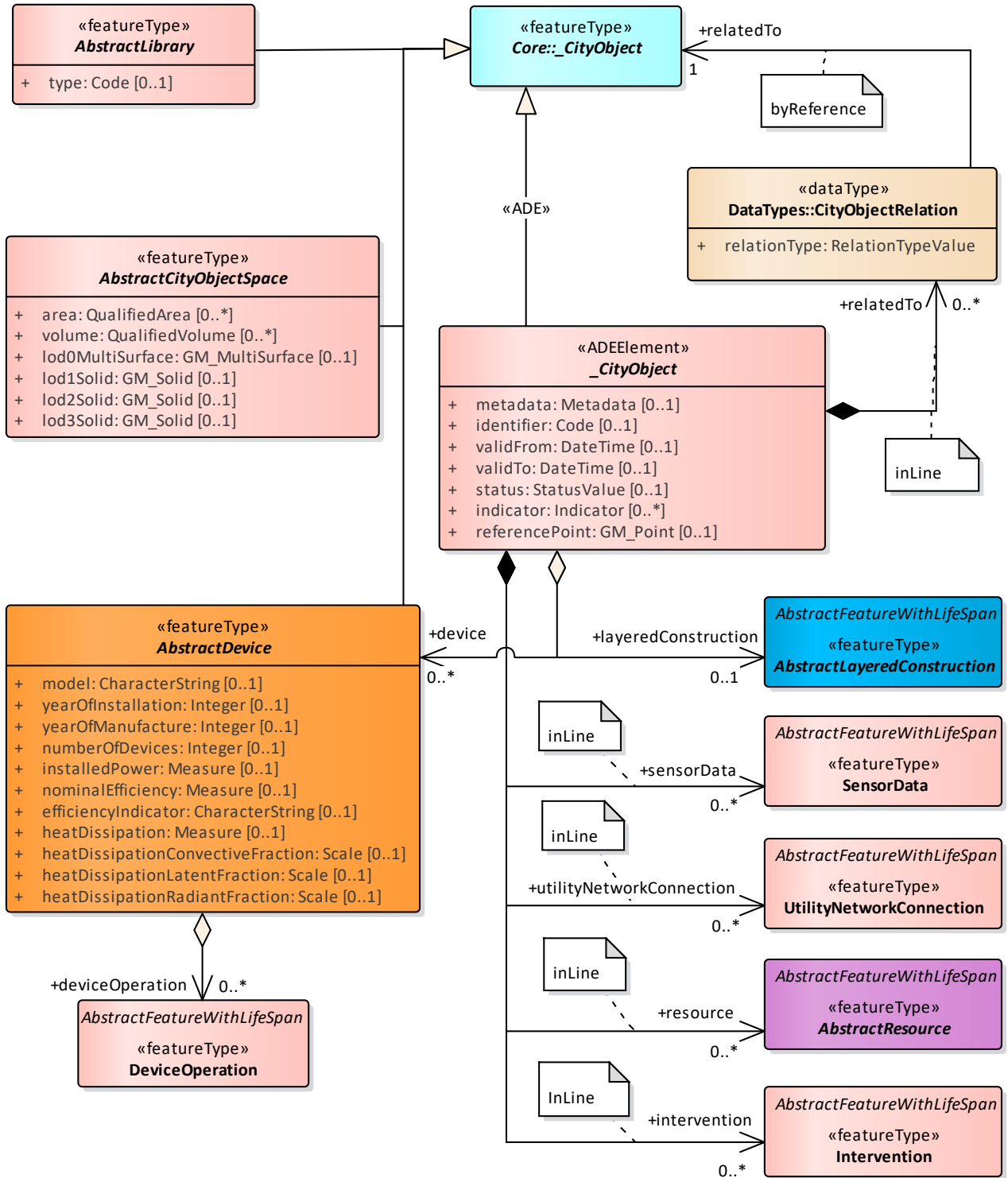
Core package part 1



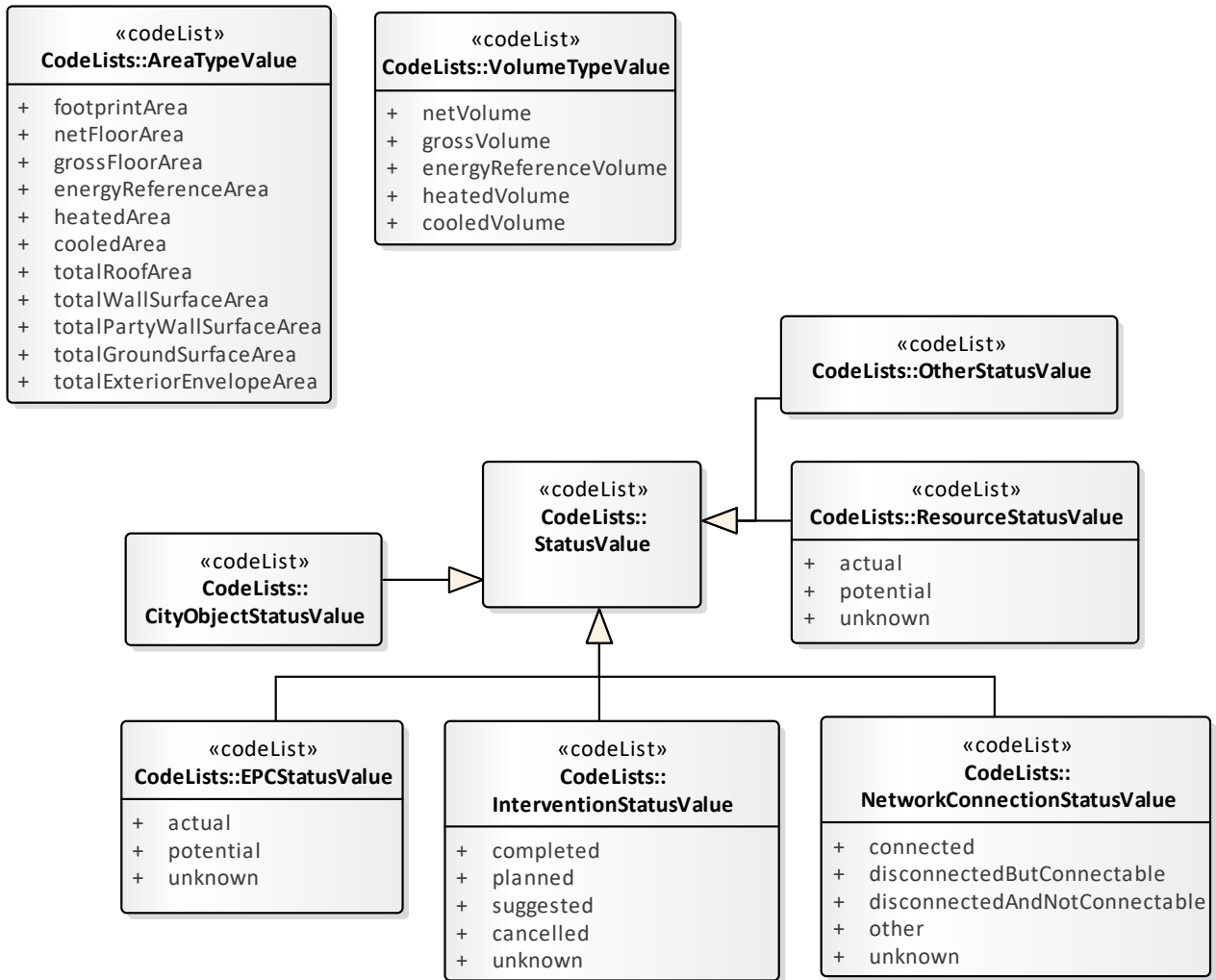
Core package part 1: CodeLists and Enumerations



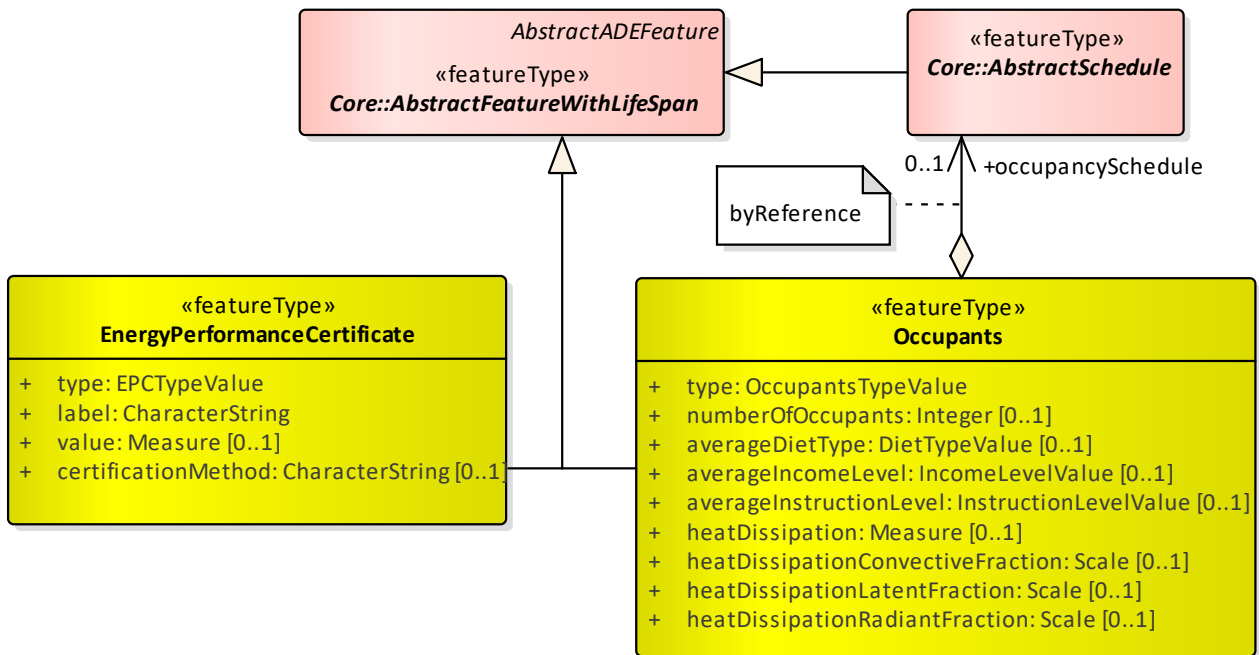
Core package part 2

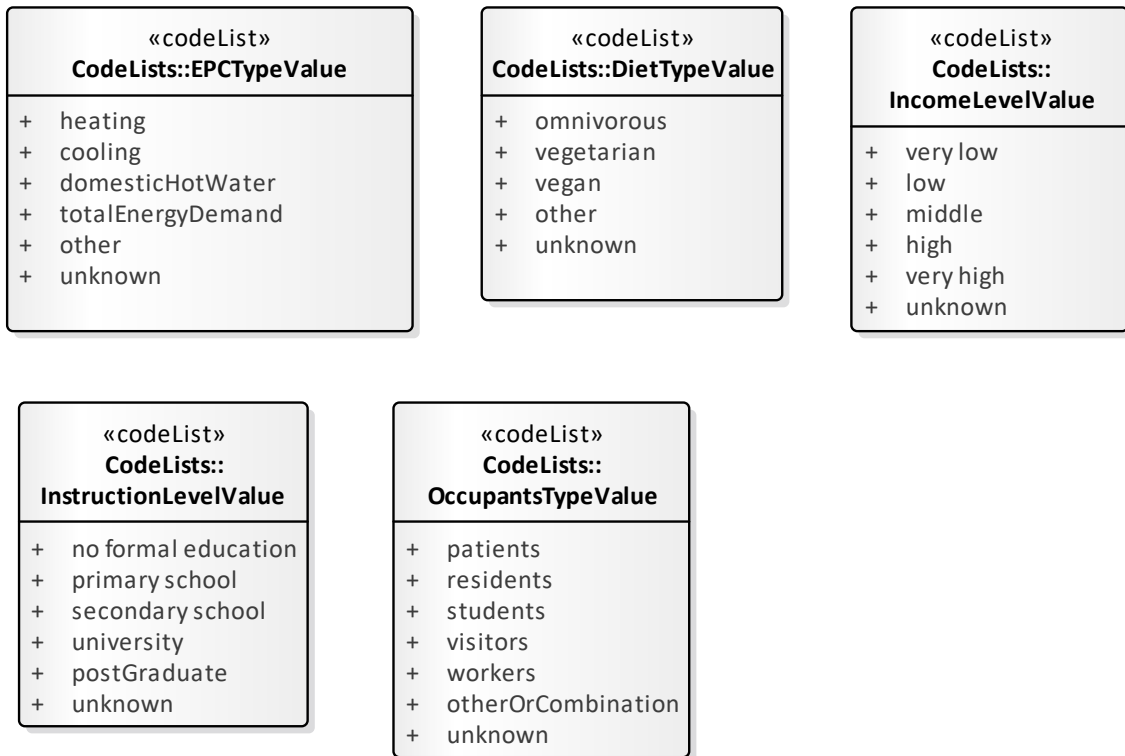


Core package part 2: CodeLists and Enumerations

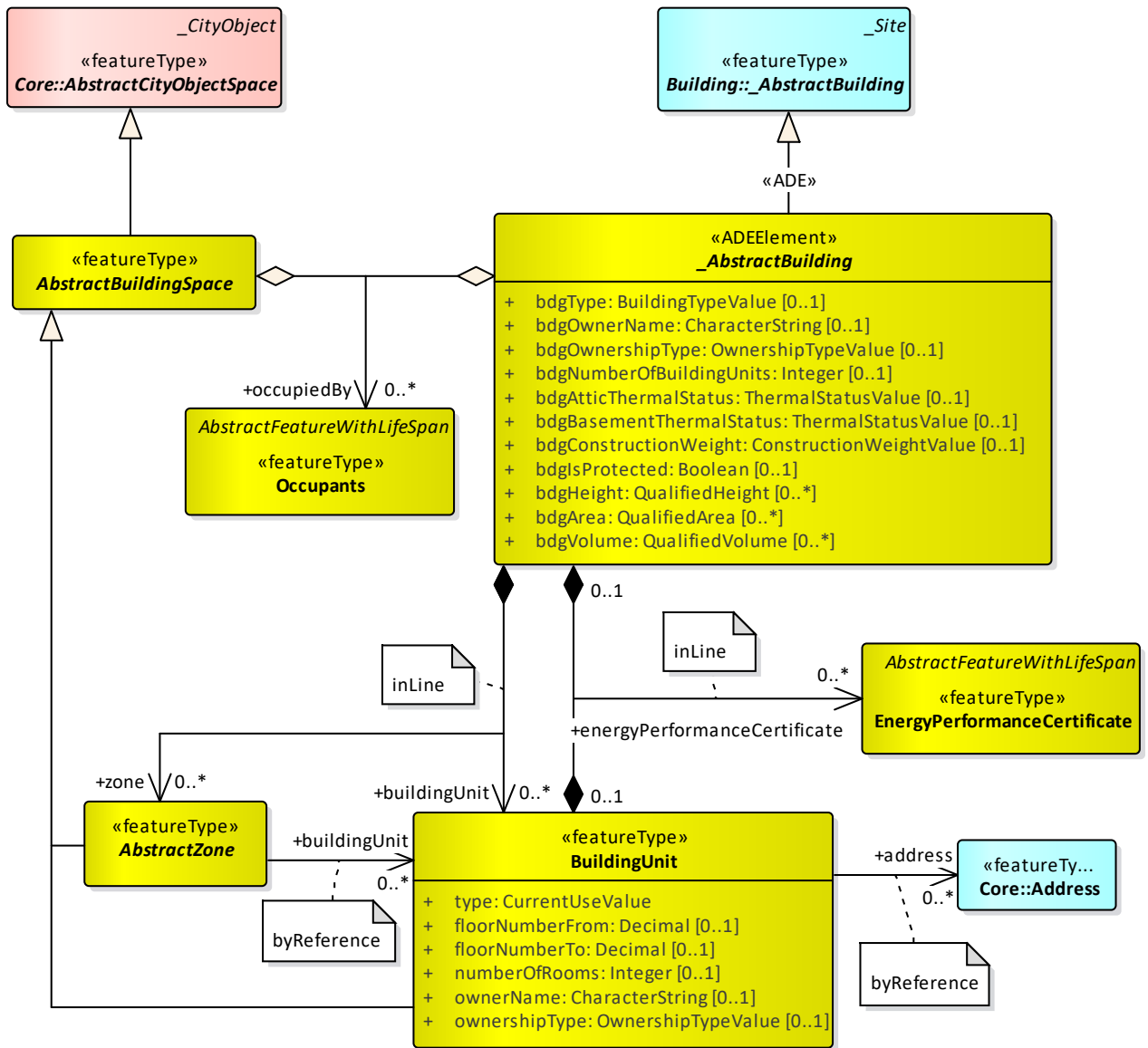


Building package part 1

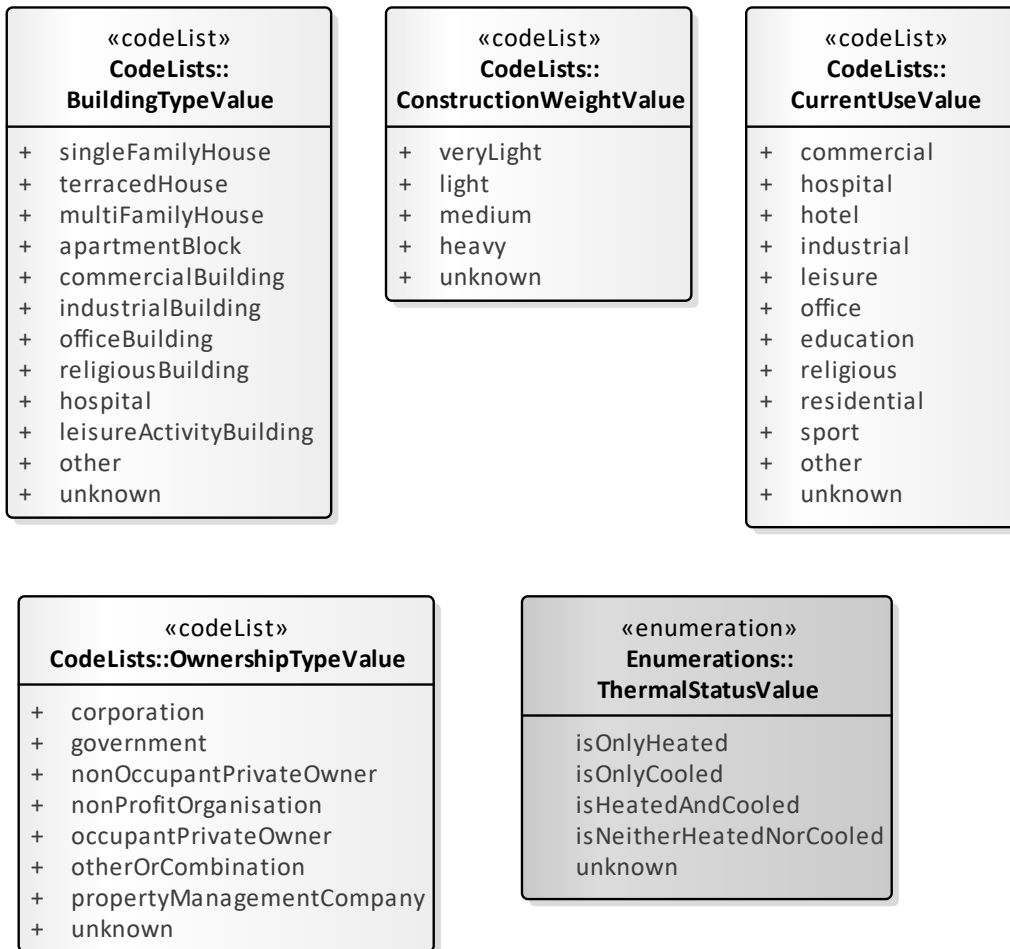


Building package part 1: CodeLists and Enumerations

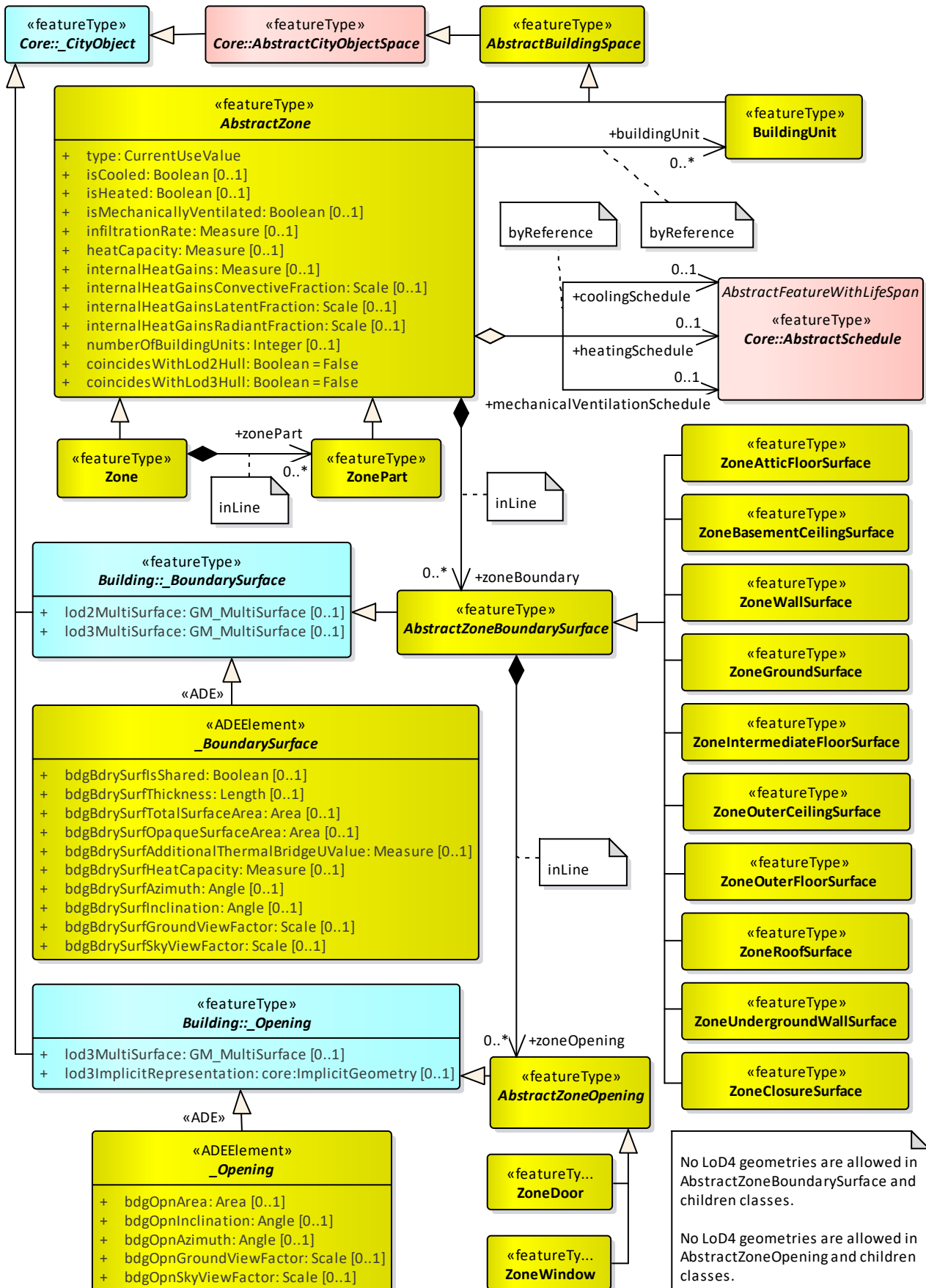
Building package part 2



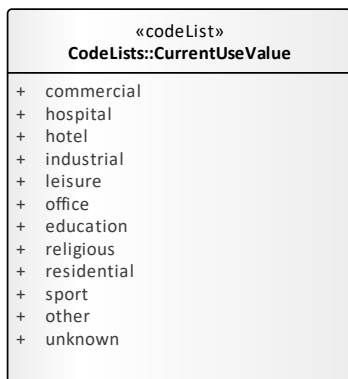
Building package part 2: CodeLists and Enumerations



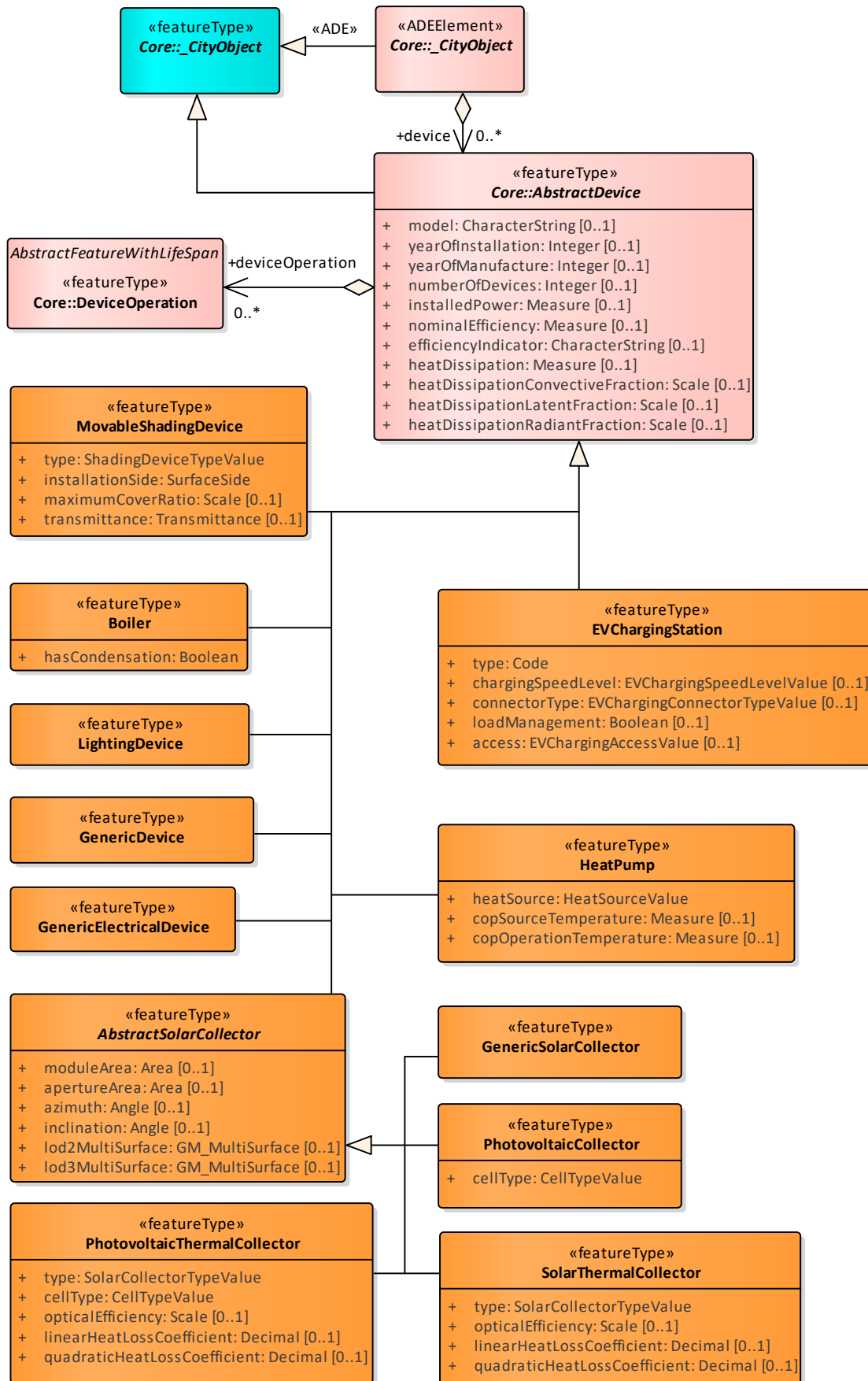
Building package part 3



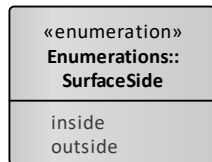
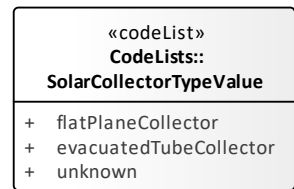
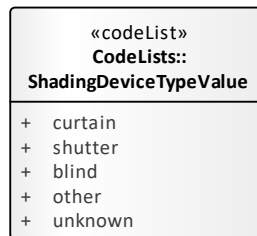
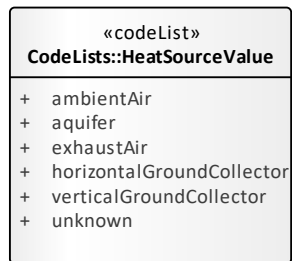
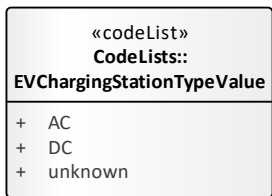
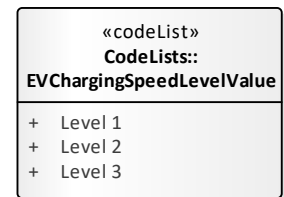
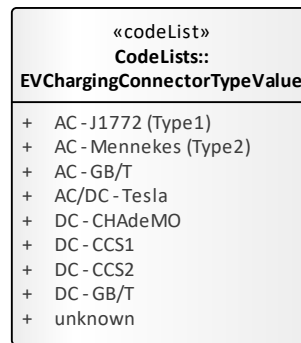
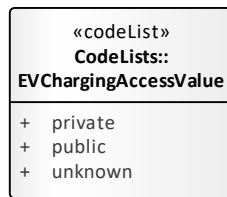
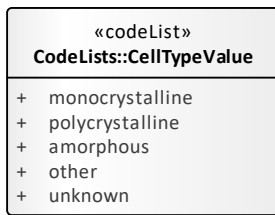
Building package part 3: CodeLists and Enumerations



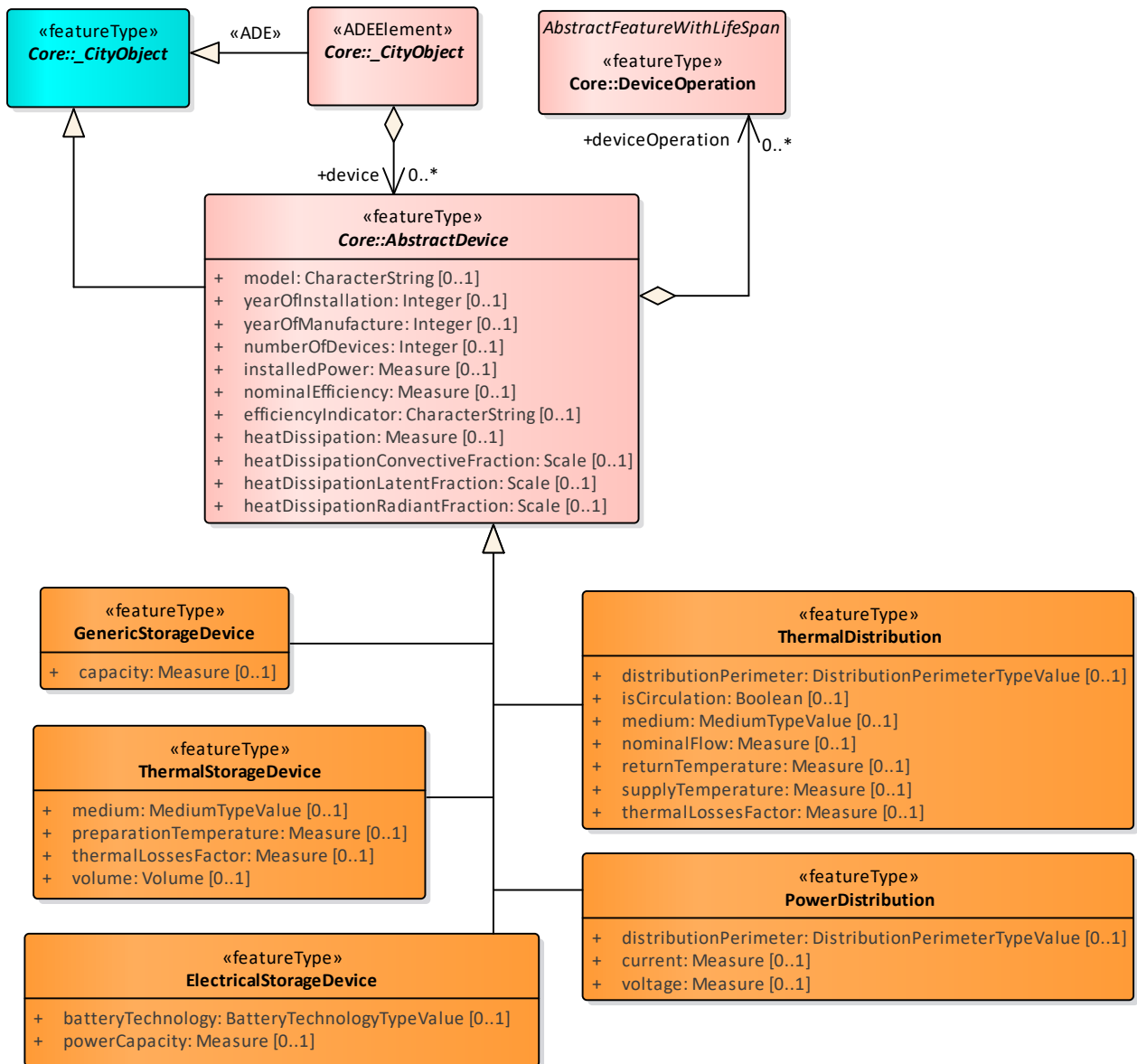
Devices package part 1



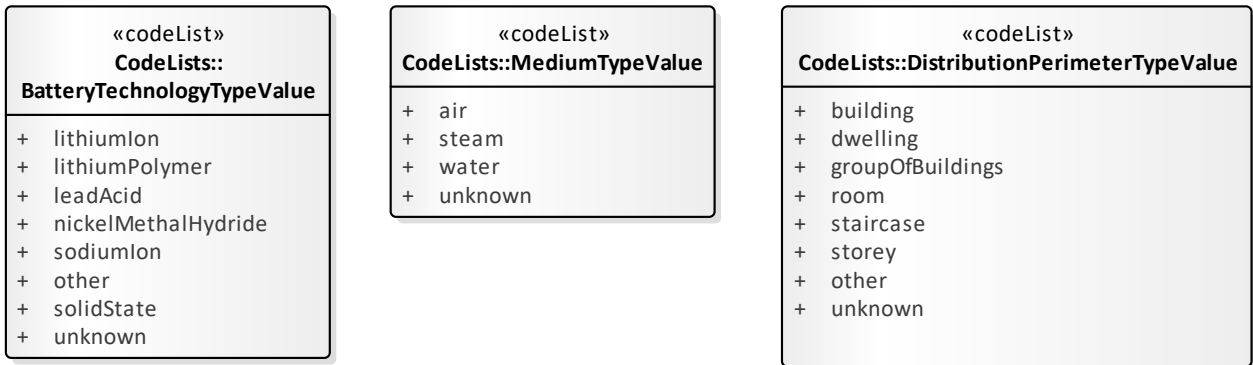
Devices package part 1: CodeLists and Enumerations



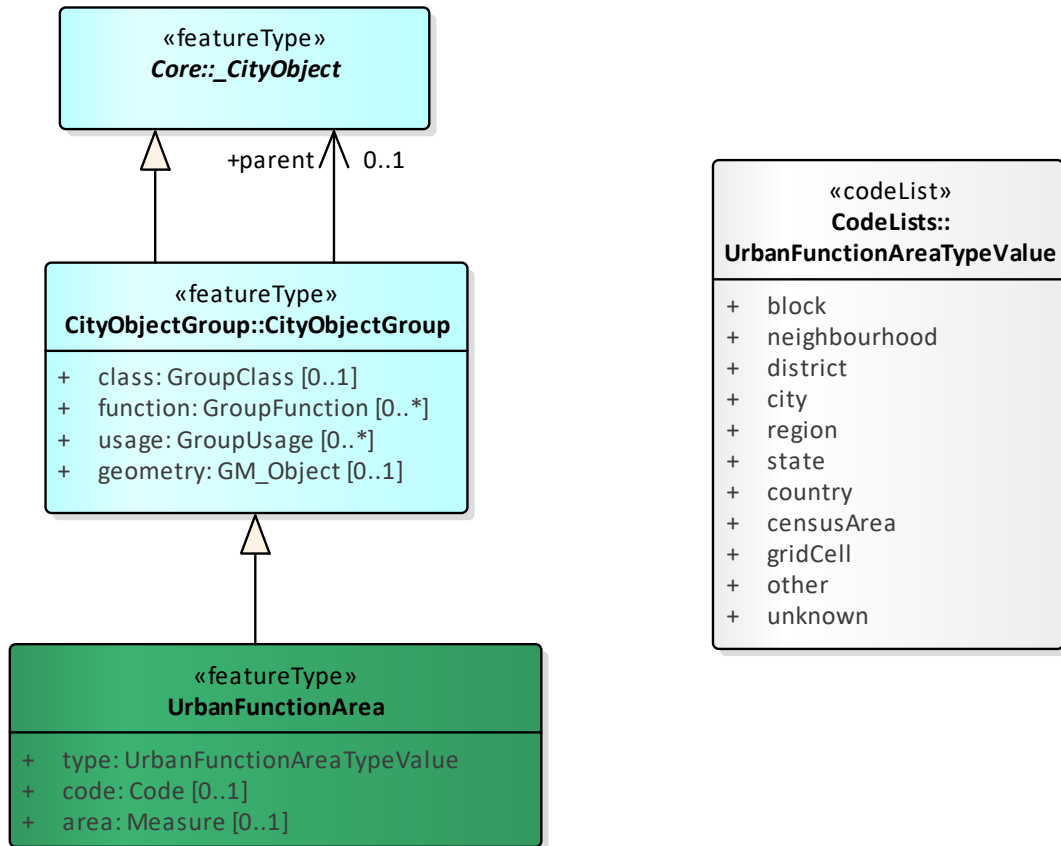
Devices package part 2



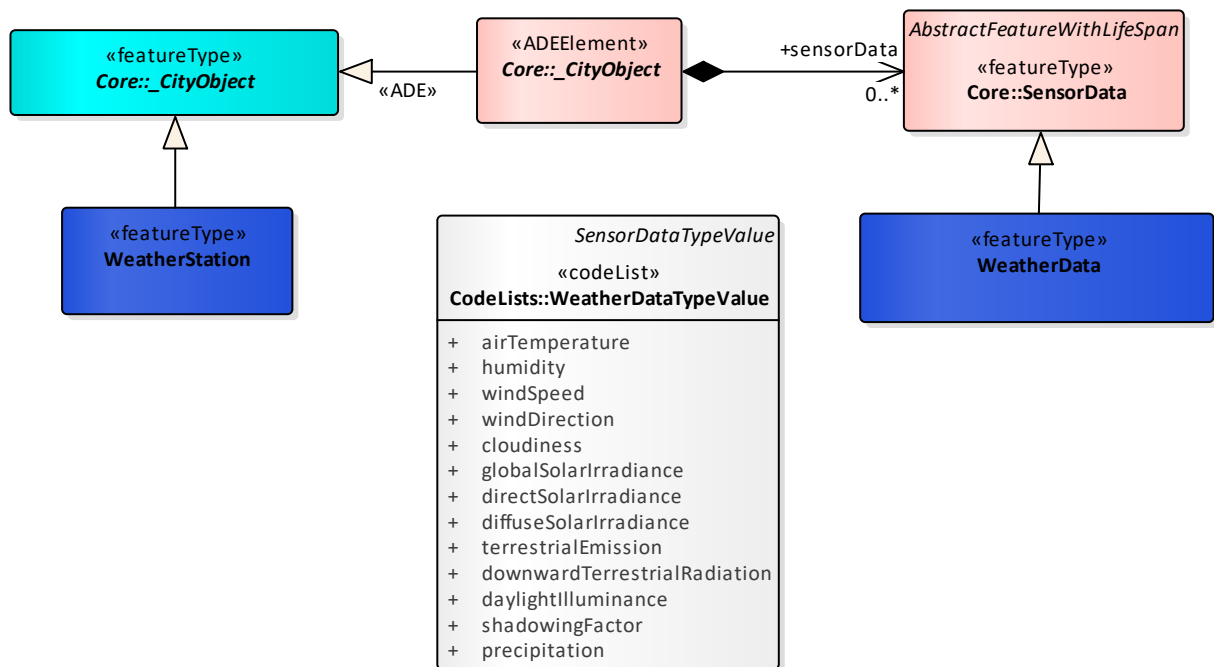
Devices package part 2: CodeLists and Enumerations



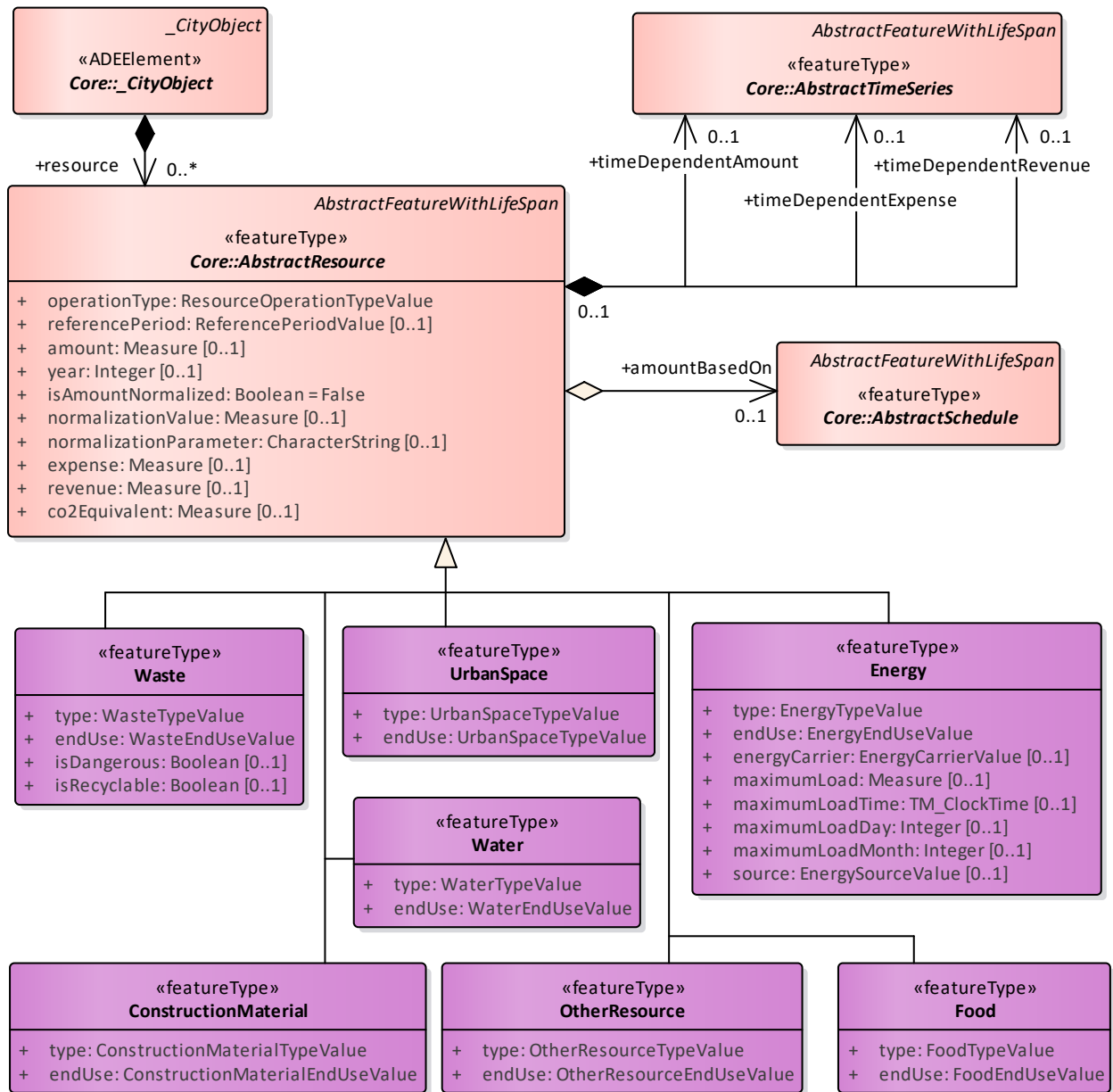
Urban function areas package and CodeLists



Weather station package and CodeLists



Resources package



Resources package: CodeLists and Enumerations

«codeList» CodeLists:: ConstructionMaterialEndUseValue
+ construction + refurbishment + otherOrCombination + unknown

«codeList» CodeLists:: ConstructionMaterialTypeValue
+ wood + tiles + concrete + sand + soil + otherOrCombination + glass + bricks + unknown

«codeList» CodeLists:: EnergyCarrierValue
+ electricity + coal + naturalGas + chilledAir + chilledWater + fuelOil + hotAir + steam + hotWater + woodChips + hydrogen + propane + woodPallets + other + unknown

«codeList» CodeLists:: EnergyEndUseValue
+ cooking + domesticHotWater + electricalAppliances + lighting + mobility + otherOrCombination + spaceCooling + spaceHeating + ventilation + process + construction + unknown

«codeList» CodeLists:: EnergyTypeValue
+ actual + net + primary + secondary + final + useful

«codeList» CodeLists:: EnergySourceValue
+ photovoltaicPanels + powerGrid + districtHeating + solarThermalPanels + heatPump + wind

«codeList» CodeLists:: FoodEndUseValue
+ alimentation + commerce

«codeList» CodeLists:: FoodTypeValue
+ greeneries + meat + cereals + cheese + milk + wine + beer + unknown

«codeList» CodeLists:: OtherResourceEndUseValue

«codeList» CodeLists:: OtherResourceTypeValue

«codeList» CodeLists:: UrbanSpaceEndUseValue
--

«codeList» CodeLists:: UrbanSpaceTypeValue
--

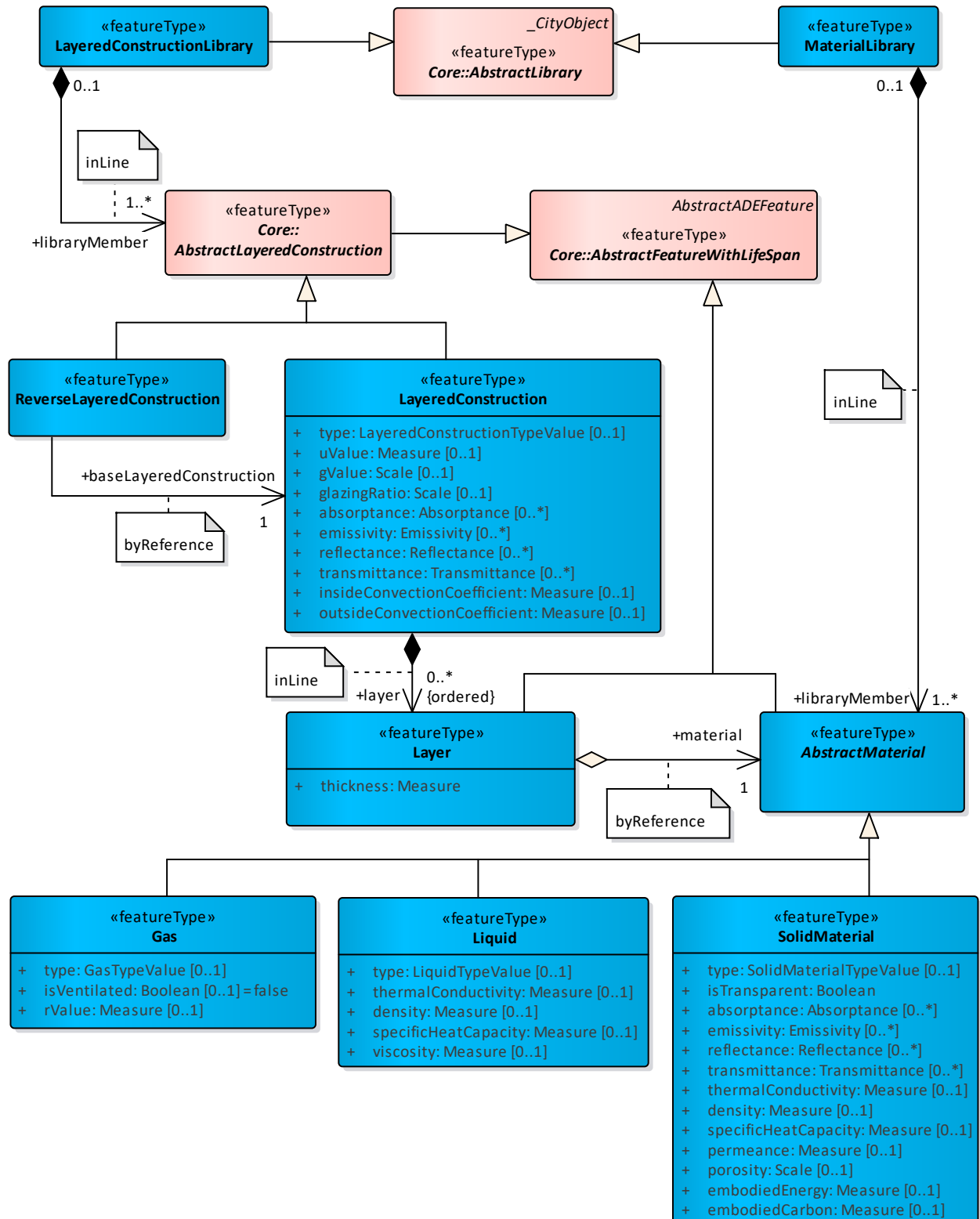
«codeList» CodeLists:: WasteEndUseValue
+ incineration + landFilling + recycling + other + unknown

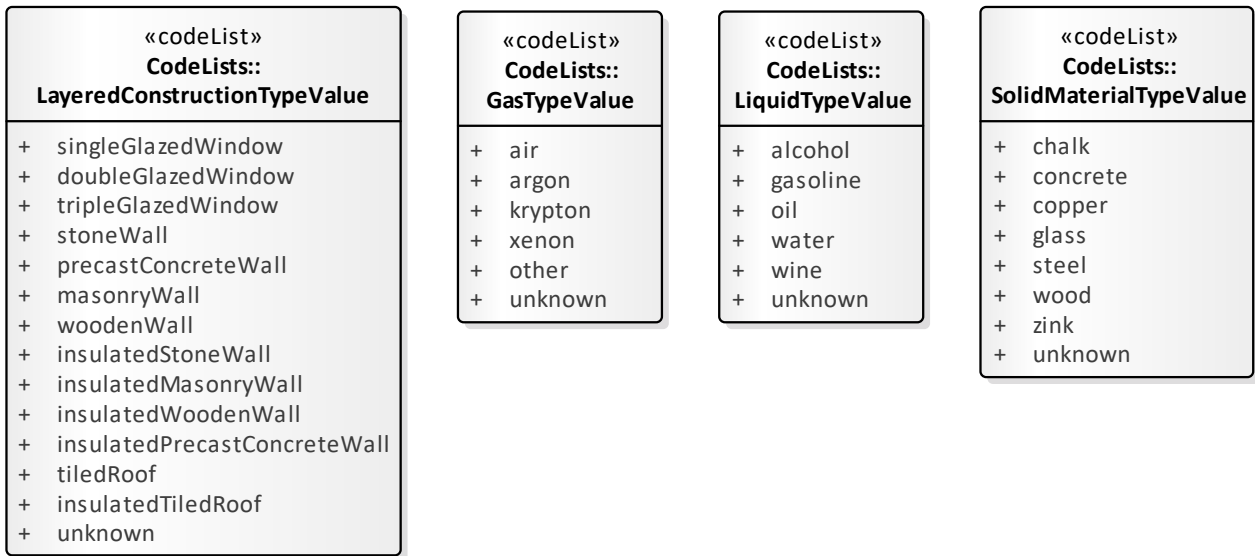
«codeList» CodeLists:: WasteTypeValue
+ organic + plastic + glass + paper + metal + water + otherOrCombination + unknown

«codeList» CodeLists:: WaterEndUseValue
+ drinking + washing + processing + otherOrCombination + unknown

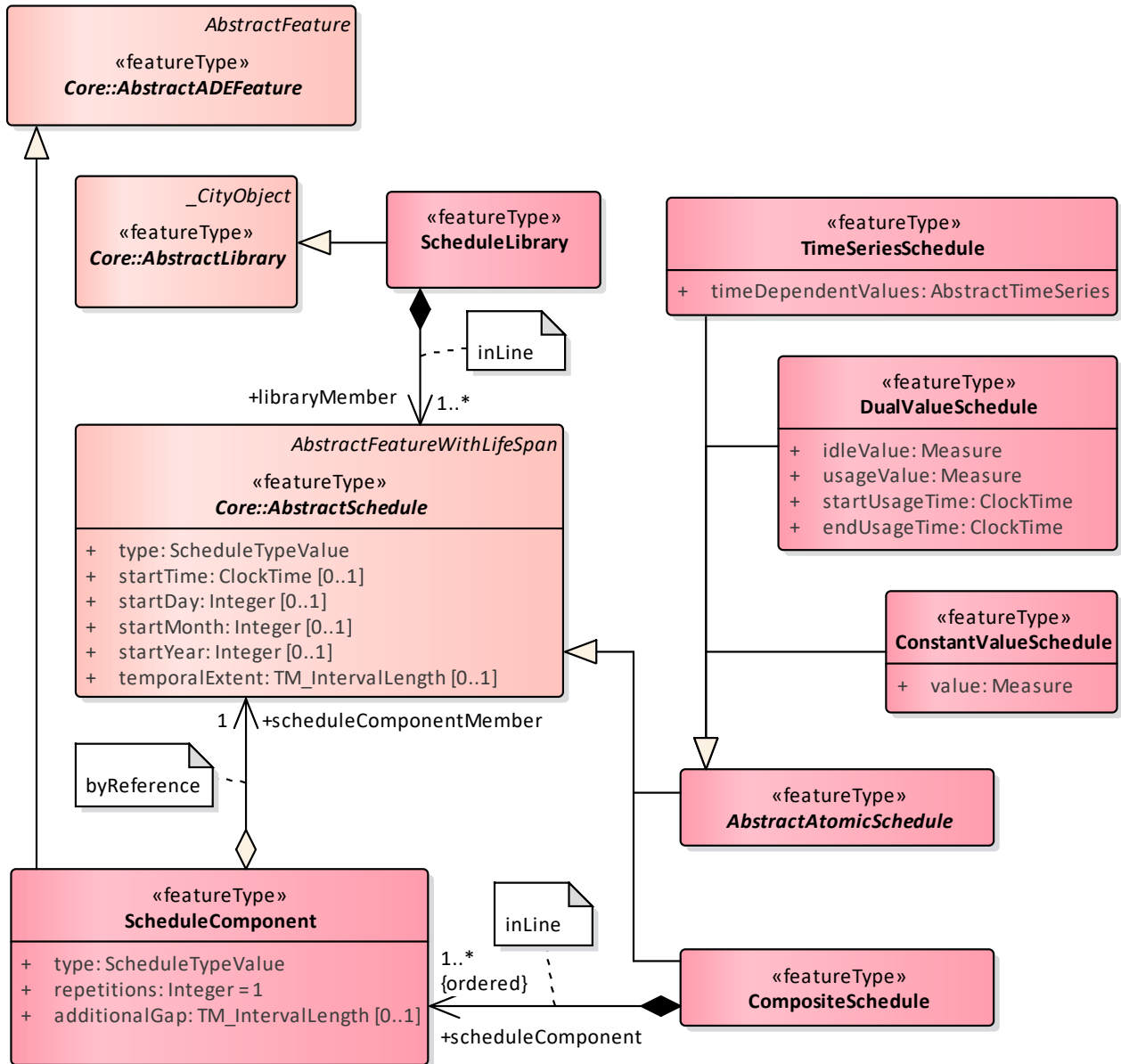
«codeList» CodeLists:: WaterTypeValue
+ drinkWater + wasteWater + rainWater + otherOrCombination + unknown

Layered construction package



Layered construction package: CodeLists

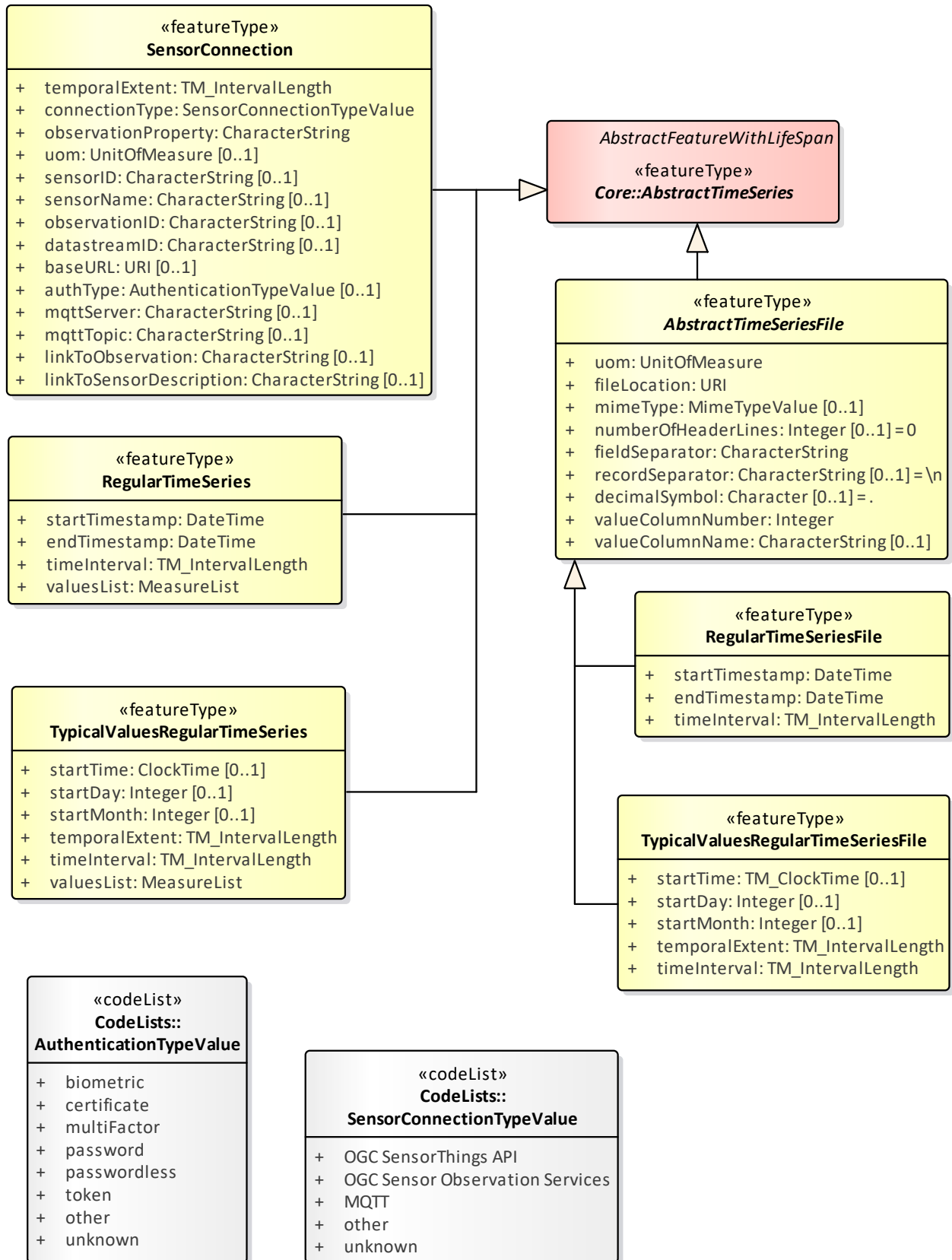
Schedules package



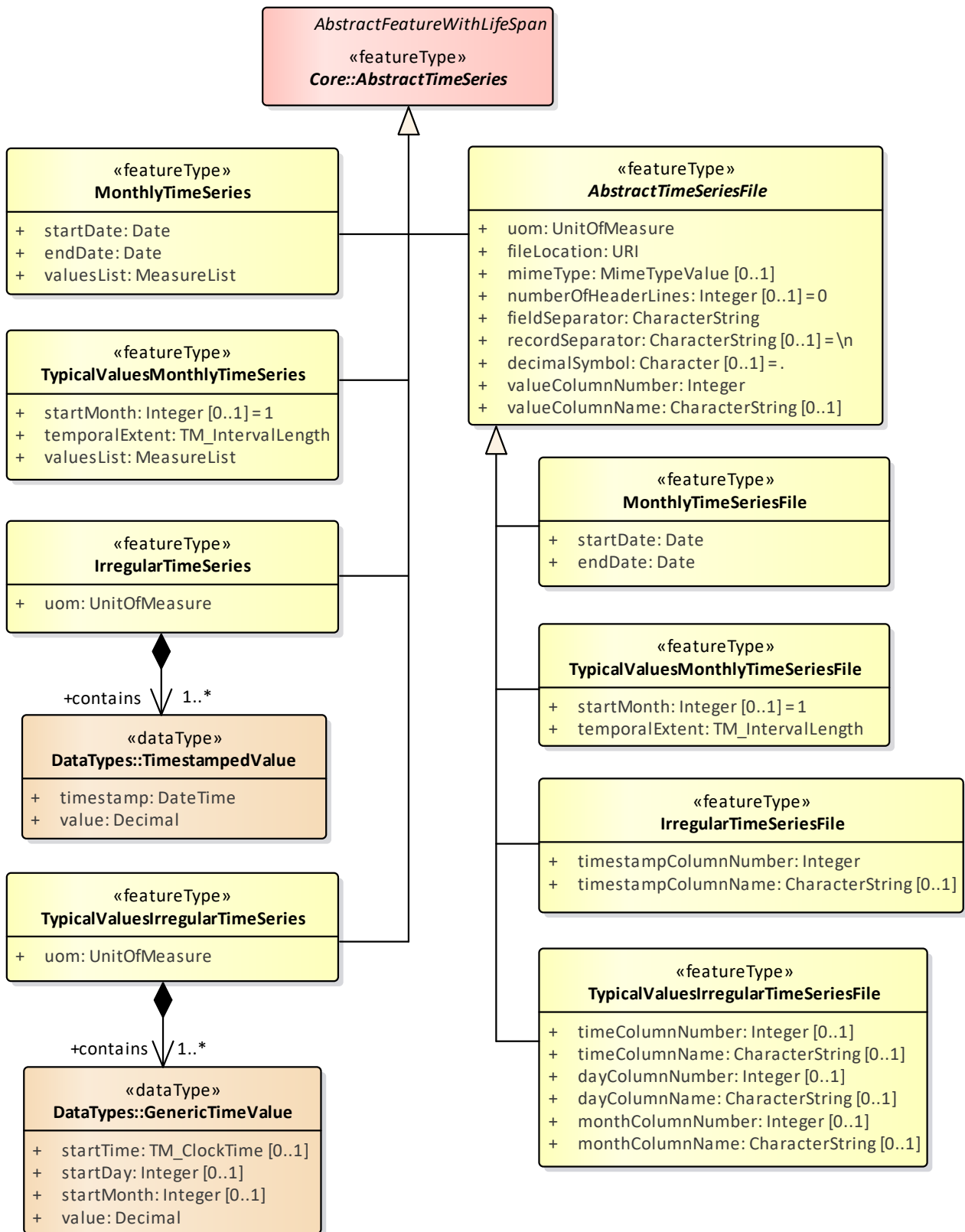
Schedules package: CodeLists

«codeList» CodeLists:: ScheduleTypeValue	
+	hour
+	day
+	week
+	month
+	season
+	year
+	typicalDay
+	typicalWeek
+	typicalMonth
+	typicalSeason
+	typicalYear
+	monday
+	tuesday
+	wednesday
+	thursday
+	friday
+	saturday
+	sunday
+	weekDay
+	weekendDay
+	beerDay
+	holiday
+	january
+	february
+	march
+	april
+	may
+	june
+	july
+	august
+	september
+	october
+	november
+	december
+	spring
+	summer
+	autumn
+	winter
+	other

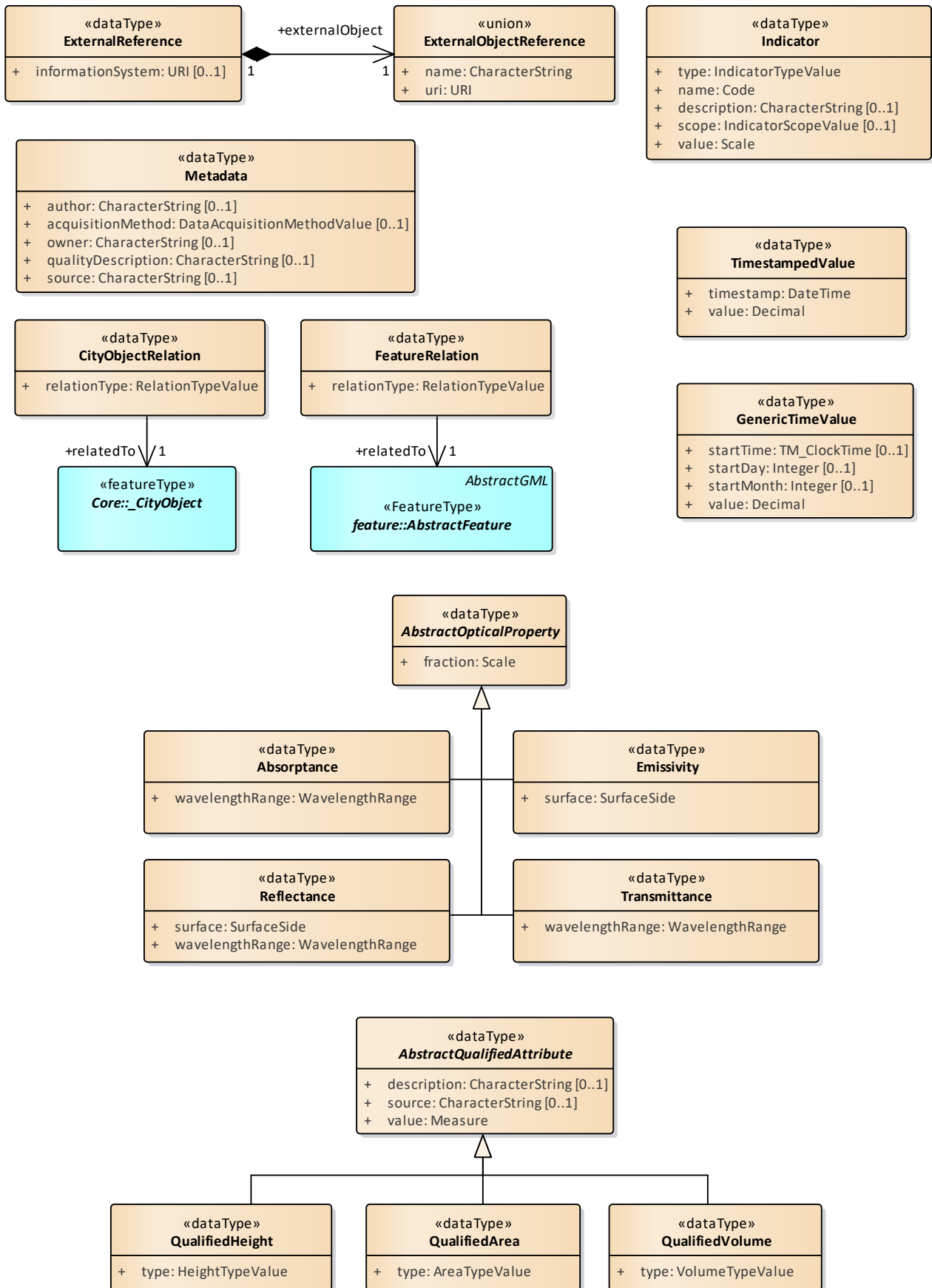
Time series package part 1 and CodeLists



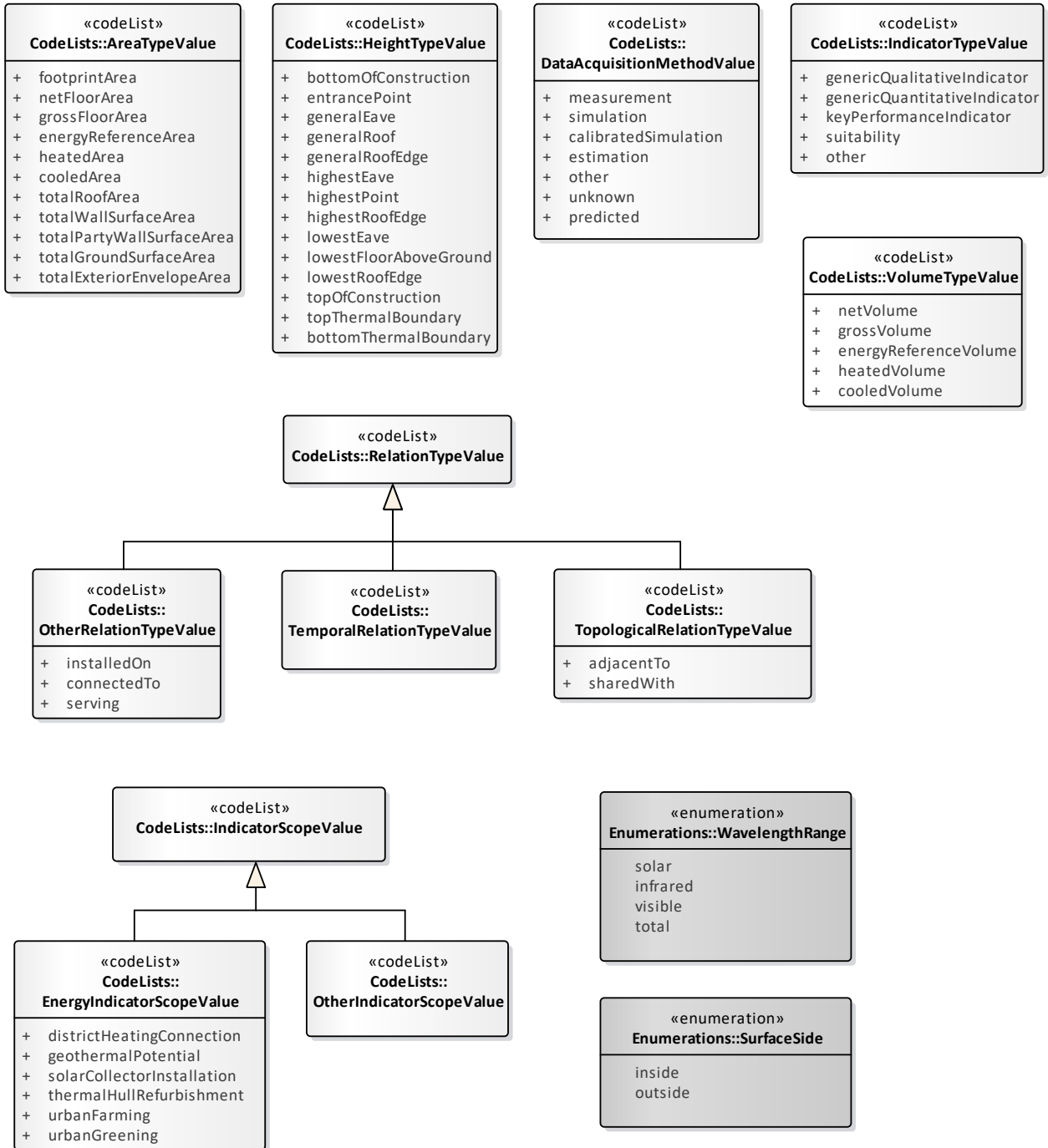
Time series package part 2



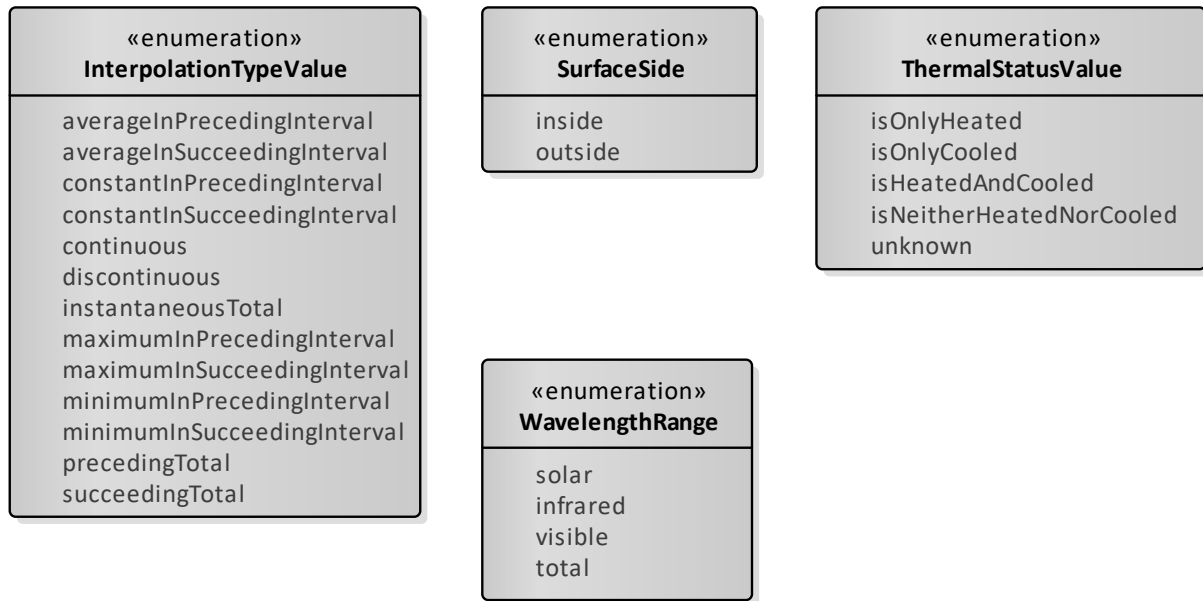
DataTypes package



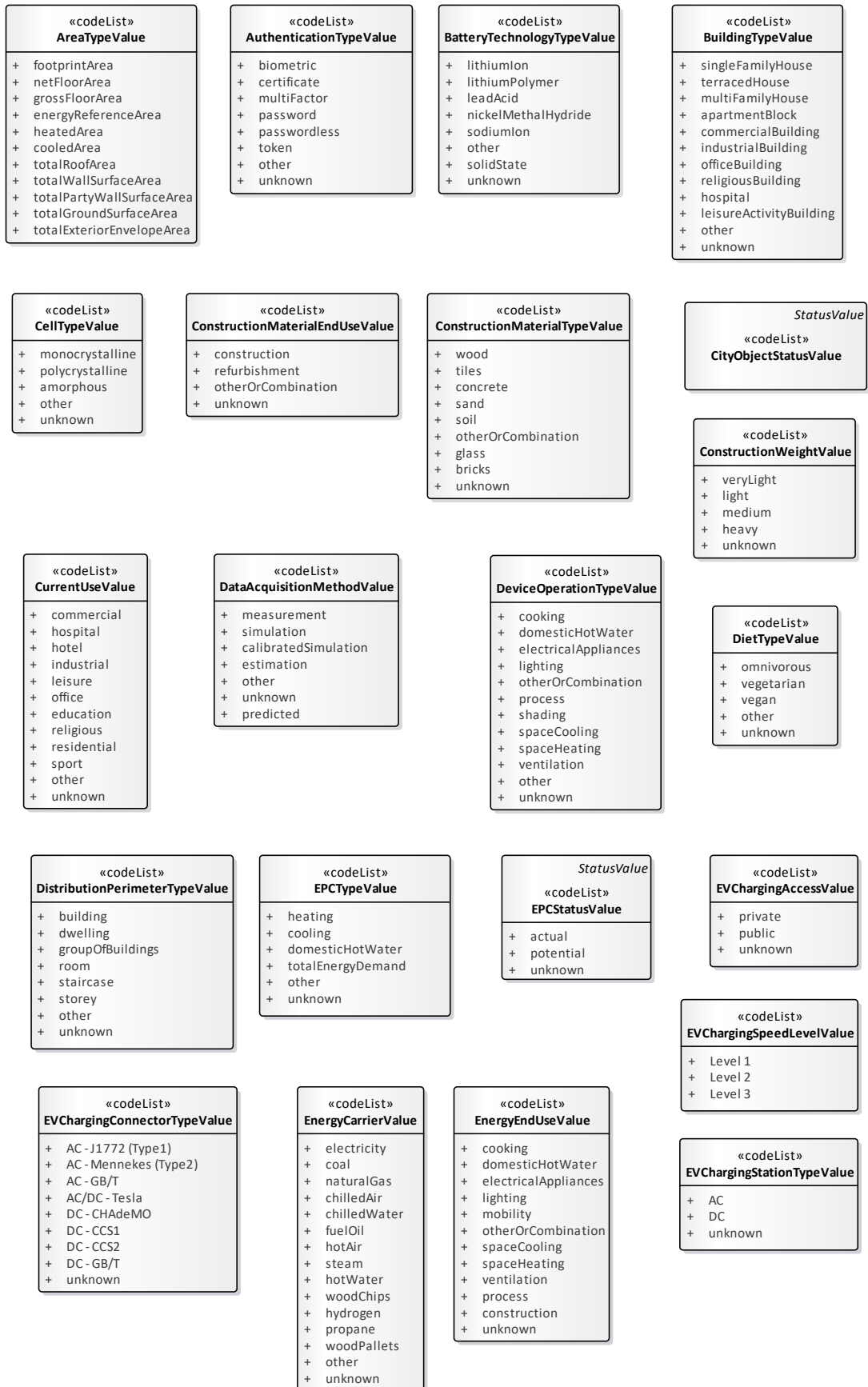
DataTypes package: CodeLists and Enumerations



Enumerations package



Codelists package



«codeList» EnergyIndicatorScopeValue
+ districtHeatingConnection
+ geothermalPotential
+ solarCollectorInstallation
+ thermalHullRefurbishment
+ urbanFarming
+ urbanGreening

«codeList» EnergySourceValue
+ photovoltaicPanels
+ powerGrid
+ districtHeating
+ solarThermalPanels
+ heatPump
+ wind

«codeList» EnergyTypeValue
+ actual
+ net
+ primary
+ secondary
+ final
+ useful

«codeList» FoodEndUseValue
+ alimentation
+ commerce

«codeList» FoodTypeValue
+ greeneries
+ meat
+ cereals
+ cheese
+ milk
+ wine
+ beer
+ unknown

«codeList» FunctionInNetworkValue
+ producing
+ consuming
+ storage
+ other
+ unknown

«codeList» GasTypeValue
+ air
+ argon
+ krypton
+ xenon
+ other
+ unknown

«codeList» HeatSourceValue
+ ambientAir
+ aquifer
+ exhaustAir
+ horizontalGroundCollector
+ verticalGroundCollector
+ unknown

«codeList» HeightTypeValue
+ bottomOfConstruction
+ entrancePoint
+ generalEave
+ generalRoof
+ generalRoofEdge
+ highestEave
+ highestPoint
+ highestRoofEdge
+ lowestEave
+ lowestFloorAboveGround
+ lowestRoofEdge
+ topOfConstruction
+ topThermalBoundary
+ bottomThermalBoundary

«codeList» IndicatorTypeValue
+ genericQualitativeIndicator
+ genericQuantitativeIndicator
+ keyPerformanceIndicator
+ suitability
+ other

«codeList» IndicatorScopeValue

«codeList» IncomeLevelValue
+ very low
+ low
+ middle
+ high
+ very high
+ unknown

«codeList» InstructionLevelValue
+ no formal education
+ primary school
+ secondary school
+ university
+ postGraduate
+ unknown

«codeList» InterventionActionValue
+ installedSolarPanels
+ installedPVPanels
+ installedHeatPump
+ changedBoiler
+ insulatedWholeThermalHull
+ installedFloorHeating
+ changedWindows
+ changedDoors
+ insulatedRoof
+ insulatedExteriorWalls
+ insulatedGroundFloor
+ other
+ unknown

StatusValue «codeList» InterventionStatusValue
+ completed
+ planned
+ suggested
+ cancelled
+ unknown

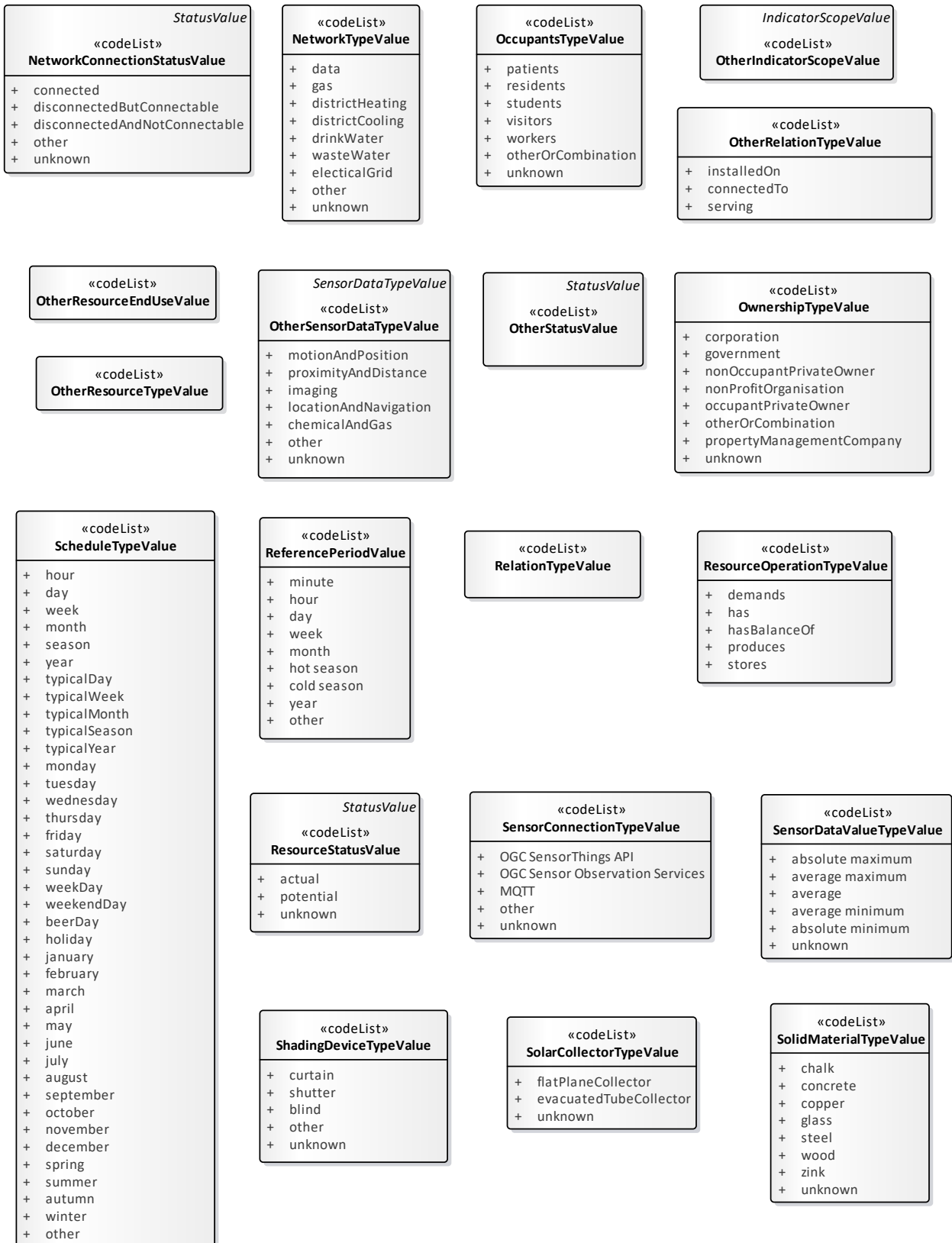
«codeList» InterventionTypeValue
+ inspection
+ maintenance
+ refurbishment
+ other
+ unknown

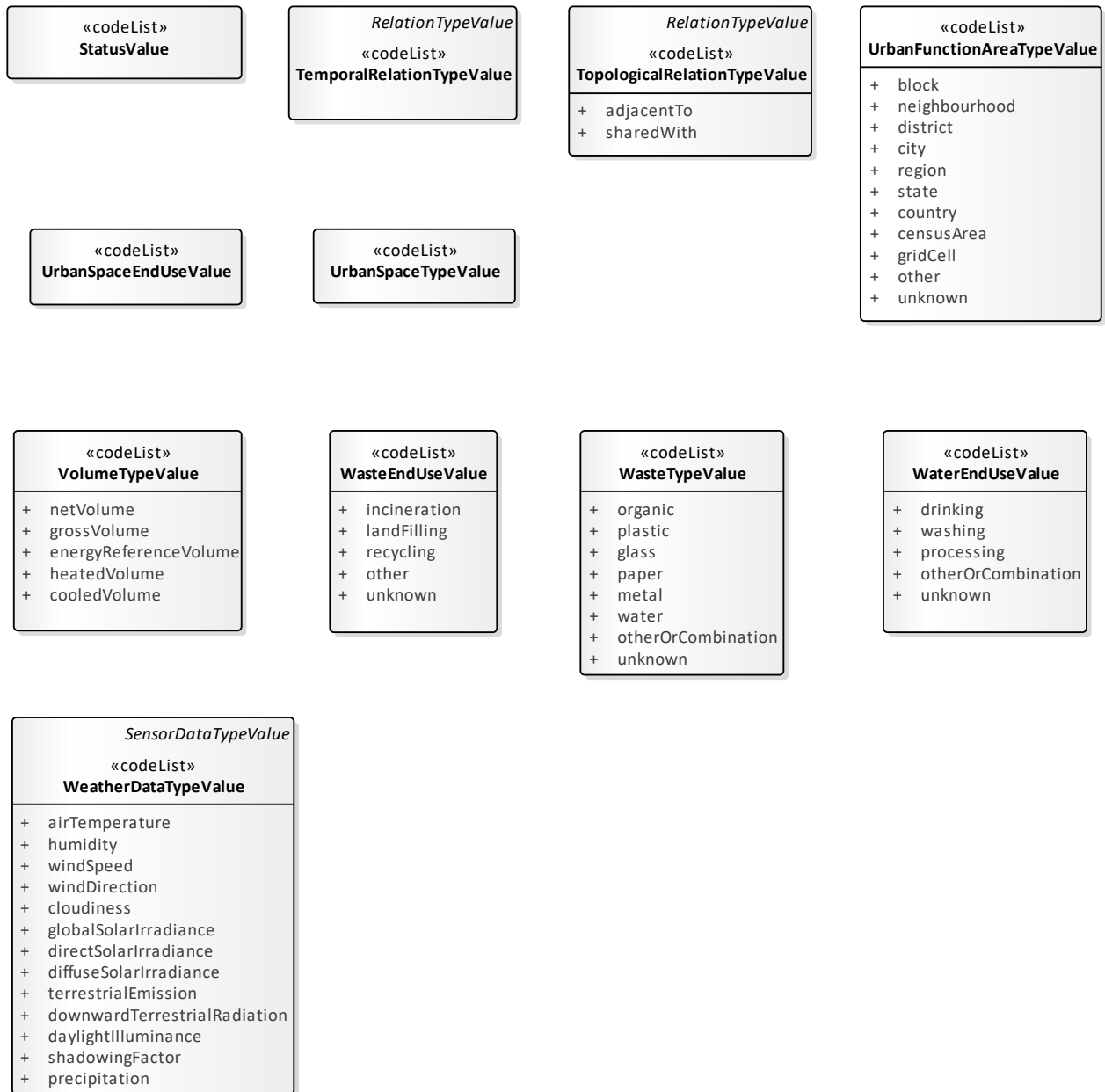
«codeList» LayeredConstructionTypeValue
+ singleGlazedWindow
+ doubleGlazedWindow
+ tripleGlazedWindow
+ stoneWall
+ precastConcreteWall
+ masonryWall
+ woodenWall
+ insulatedStoneWall
+ insulatedMasonryWall
+ insulatedWoodenWall
+ insulatedPrecastConcreteWall
+ tiledRoof
+ insulatedTiledRoof
+ unknown

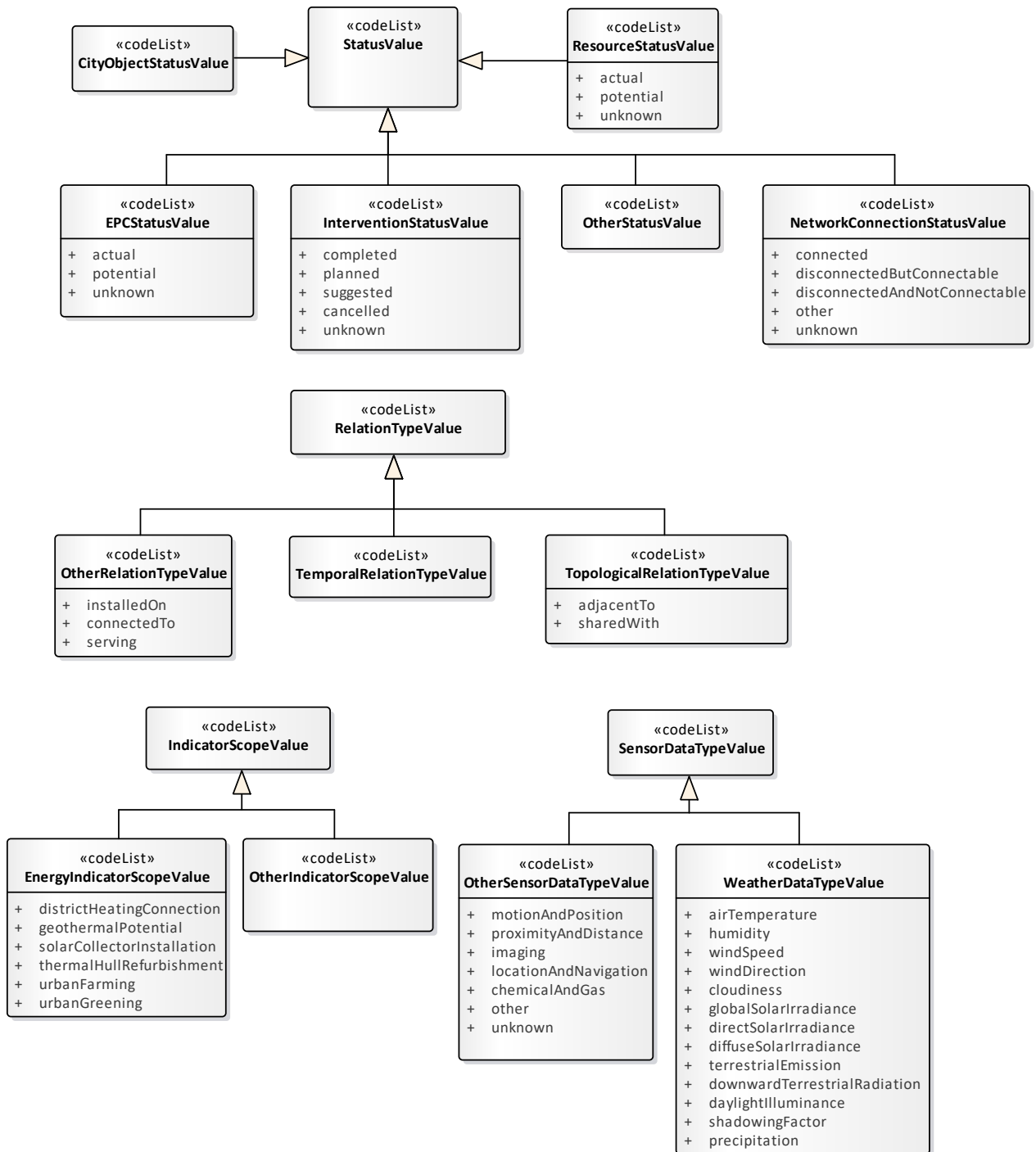
«codeList» LiquidTypeValue
+ alcohol
+ gasoline
+ oil
+ water
+ wine
+ unknown

«codeList» MediumTypeValue
+ air
+ steam
+ water
+ unknown

«codeList» MimeTypeValue







Changelog

Version 3.0 beta 8

Cumulative list of changes since beta 7 (as of 2026-03-22)

CodeLists package:

Several CodeList classes added and some renamed. Added values to several CodeLists.

Enumerations package:

- Class **NetworkConnectionStatusValue** has been changed to a CodeList and moved to the corresponding package.

DataTypes package:

- Dropped class **Suitability**
- Added class **Absorptance**
- Added class **CityObjectRelation**
- Added class **ExternalObjectReference**
- Added class **ExternalReference**
- Added class **FeatureRelation**
- Added class **GenericTimeValue**
- Added class **Indicator**
- Added class **Metadata**
- Added class **TimestampedValue**

Core package:

- Dropped class **AbstractBuildingUnit**
- Dropped class **AbstractUsageZone**
- Dropped class **AbstractThermalZone**
- Dropped class **RefurbishmentMeasure**
- Moved class **EnergyPerformanceCertification** to Building package
- Moved class **ADE _AbstractBuilding** to Building package
- Renamed class **AbstractBuildingPartition** to **AbstractCityObjectSpace**
- Renamed class **WeatherData** to **SensorData**
- Added class **AbstractADEFeature**
- Added class **AbstractFeatureWithLifeSpan**
- Added class **Intervention**
- Classes **AbstractResource**, **AbstractSchedule**, **AbstractLayeredConstruction**, **UtilityNetworkCnnection**, **SensorData**, **DeviceOperation** are now derived from class **AbstractFeatureWithLifeSpan**
- Class **ADE _CityObject**:
 - Dropped property *suitability*
 - Renamed property from *weatherData* to *sensorData*
 - Added property *metadata*
 - Added property *identifier*
 - Added property *validFrom*
 - Added property *validTo*
 - Added property *status*
 - Added property *indicator*
- Class **AbstractLibrary**:
 - Dropped property *source*
 - Dropped property *author*
- Class **AbstractCityObjectSpace**:
 - Added property *lod0MultiSurface*
- Class **AbstractSchedule**:
 - Dropped property *libraryCode*
- Class **AbstractResource**:
 - Dropped property *status*
 - Dropped property *amountType*

- Renamed property *costsMoney* to *expense*
- Renamed property *yieldsMoney* to *revenue*
- Added property *referencePeriod*
- Added property *amountBasedOn*
- Added property *timeDependentExpense*
- Added property *timeDependentRevenue*
- Class **AbstractTimeSeries**:
 - Dropped property *acquisitionMethod*
 - Dropped property *source*

Building physics package:

Dropped. Some classes reworked and moved to new module Building

Occupancy package:

Dropped. Some classes reworked and moved to new module Building

Building package:

This module is new. It reworks and merges the previous Building physics module and Occupancy module. Additionally, it incorporates also class *ADE_AbstractBuilding* and its subclasses.

- Class **ADE_AbstractBuilding**
 - Class is moved from Core module
 - Dropped property *usageZone*
 - Dropped property *thermalZone*
 - Added property *occupiedBy*
 - Added property *zone*
 - Added property *owner*
 - Added property *ownershipType*
 - Added property *numberOfBuildingUnits*
- Class **EnergyPerformanceCertificate**
 - Class is moved from Core module
 - Class is derived from class **AbstractADEFeatureWithLifeSpan**
 - Dropped property *type*
 - Dropped property *issueDate*
 - Dropped property *expirationDate*
 - Dropped property *certificationUri*
- Class **Occupants**
 - Class is taken over from former Occupancy module
 - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
 - Renamed relation *occupancyRate* to *occupancySchedule*
- Added class **AbstractBuildingSpace**
- Class **BuildingUnit**
 - Class is taken over from former Occupancy module
 - Class is now derived from class **AbstractBuildingSpace**
 - Added property *floorNumberFrom*
 - Added property *floorNumberTo*
- Added class **AbstractZone**
- Added class **Zone**
- Added class **ZonePart**
- Class **ADE_BoundarySurface**
 - Class is taken over from former Building physics module
 - Dropped property *bdgBdrySurfIsAdiabatic*
 - Added property *bdgBdrySurfIsShared*
 - Added property *bdgBdryAdditionalThermalBridgeUValue*
- Added class **AbstractZoneBoundarySurface**
- Added class **ZoneAtticFloorSurface**
- Added class **ZoneBasementCeilingSurface**
- Added class **ZoneExteriorWallSurface**
- Added class **ZoneGroundSurfaceSurface**

- Added class **ZoneIntermediateFloorSurface**
- Added class **ZoneRoofSurface**
- Added class **ZoneUndergrounSurface**
- Added class **ZoneClosureSurface**
- Added class **ADE _Opening**
 - Class is taken over from former Building physics module
- Added class **ADE AbstractZoneOpening**
- Added class **ZoneDoor**
- Added class **ZoneWindow**

Devices package:

- Added class **GenericStorageDevice**
- Added class **EVChargingStation**
- Added class **ThermalDistribution**
- Added class **PowerDistribution**
- Class **AbstractDevice**
 - Added property *yearOfInstallation*

Urban function area package

- Class **UrbanFunctionArea**
 - Added property *area*
 - Set type of property *code* to gml:Code

WeatherStation package

- Added class **WeatherData**

Resources package

- Class **Energy**
 - Added property *maximulLoad*
 - Added property *maximulLoadTime*
 - Added property *maximulLoadDay*
 - Added property *maximulLoadMonth*

Layered Construction package

- Class **MaterialLibrary**
 - Changed multiplicity of property *libraryMember* to 1..*
- Class **LayeredConstructionLibrary**
 - Changed multiplicity of property *libraryMember* to 1..*
- Class **LayeredConstructionMaterial**
 - Dropped property *libraryCode*
 - Added property *type*
 - Added property *absorptance*
 - Added property *insideConvectionCoefficient*
 - Added property *outsideConvectionCoefficient*
- Class **AbstractMaterial**
 - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
 - Dropped property *libraryCode*
- Class **Layer**
 - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
- Class **SolidMaterial**
 - Added property *type*
 - Added property *isTransparent*
 - Added property *absorptance*
 - Added property *emissivity*
 - Added property *reflectance*
 - Added property *transmittance*
- Added class **Liquid**
- Class **Gas**
 - Added property *type*

Schedules package:

- Dropped class **AtomicSchedule**
- Dropped class **AtomicScheduleValue**
- Added class **AbstractAtomicSchedule**
- Added class **ConstantValueSchedule**
- Added class **TimeSeriesSchedule**
- Class **DualValueSchedule** is now derived from **AbstractAtomicSchedule**
- Class **ScheduleComponent** is now derived from **AbstractADEFeatureWithLifeSpan**

Time series package:

- Added class **AbstractTimeSeriesFile**
- Added class **MonthlyTimeSeries**
- Added class **MonthlyTimeSeriesFile**
- Added class **TypicalValuesMonthlyTimeSeries**
- Added class **TypicalValuesMonthlyTimeSeriesFile**
- Added class **IrregularTimeSeries**
- Added class **IrregularTimeSeriesFile**
- Added class **TypicalValuesIrregularTimeSeries**
- Added class **TypicalValuesIrregularTimeSeriesFile**
- Renamed class **TypicalValuesTimeSeries** to **TypicalValuesRegularTimeSeries**
- Renamed class **TypicalValuesTimeSeriesFile** to **TypicalValuesRegularTimeSeriesFile**
- Class **RegularTimeSeries**:
 - Dropped property *temporalExtent*
 - Added property *startTimestamp*
 - Added property *endTimestamp*
- Class **RegularTimeSeriesFile**:
 - Class is now derived from class **AbstractTimeSeriesFile**
 - Dropped property *temporalExtent*
 - Dropped property *uom*
 - Dropped property *fileURI*
 - Dropped property *numberOfHeaderLines*
 - Dropped property *fieldSeparator*
 - Dropped property *recordSeparator*
 - Dropped property *decimalSymbol*
 - Dropped property *valueColumnNumber*
 - Added property *startTimestamp*
 - Added property *endTimestamp*
- Class **TypicalValuesRegularTimeSeriesFile**:
 - Class is now derived from class **AbstractTimeSeriesFile**
 - Dropped property *uom*
 - Dropped property *fileURI*
 - Dropped property *numberOfHeaderLines*
 - Dropped property *fieldSeparator*
 - Dropped property *recordSeparator*
 - Dropped property *decimalSymbol*
 - Dropped property *valueColumnNumber*

Version 3.0 beta 7

2025-10-14

No changes, except for version number going from 2.0 to 3.0.

Version 2.0 beta 7

2025-06-25**Schedules package:**

- Class **ScheduleLibrary**
 - Changed relation type of *libraryMember* to composition

Layered Construction package:

- Class **LayeredConstructionLibrary**
 - Changed relation type of *libraryMember* to composition
- Class **LayeredConstruction**
 - Added property *gValue*
- Class **MaterialLibrary**
 - Changed relation type of *libraryMember* to composition

2025-05-30**Schedules package:**

- Class **AtomicSchedule**
 - Changed multiplicity of relation *timeSeries* to [0..1]
 - Added property *constantValue*

2025-05-13**Building physics package:**

- Class **ThermalZone**
 - Changed relation type of *thermalBoundary* to composition

2025-05-07**Core package:**

- Class **ADE_CityObject**
 - Changed relation type of *device* to aggregation
 - Changed relation type of *weatherData* to composition
- Class **AbstractThermalZone**
 - Changed relation type of *usageZone* to association
- Class **AbstractUsageZone**
 - Changed relation type of *buildingUnit* to association
- Class **AbstractResource**
 - Made relation to **AbstractTimeSeries** explicit and set to composition
- Class **DeviceOperation**
 - Made relation to **AbstractSchedule** explicit and set to aggregation
- Class **WeatherData**
 - Made relation to **AbstractTimeSeries** explicit and set to composition

Occupancy package:

- Class **UsageZone**
 - Changed relation type of *occupiedBy* to aggregation
 - Made all relations to **AbstractSchedule** explicit and set to aggregation
- Class **BuildingUnit**
 - Changed relation type of *occupiedBy* to aggregation
- Class **Occupants**
 - Made relation to **AbstractSchedule** explicit and set to aggregation

Layered Construction package:

- Class **LayeredConstruction**
 - Changed relation type of *layer* to composition
- Class **Layer**
 - Changed relation type of *material* to aggregation

Schedules package:

- Class **ScheduleComponent**
 - Changed relation type of *scheduleComponentMember* to aggregation

2025-05-01**Core package:**

- Class **ADE _AbstractSchedule**
 - Moved from package Schedules to Core
- Class **ADE _AbstractTimeSeries**
 - Moved from package Time Series to Core

Occupancy package:

- Class **Occupants**:
 - Added property *averageDietType*
 - Added property *averageIncomeLevel*
 - Added property *averageInstructionLevel*

CodeLists package:

- Added class **DietTypeValue**
- Added class **InstructionLevelValue**
- Added class **IncomeLevelValue**

2025-04-23**CodeLists package:**

- Added values to class **AreaTypeValue**
- Renamed class **HeightReferenceValue** to **HeightTypeValue**
- Added class **SuitabilityReasonValue**
- Added class **UrbanSpaceTypeValue**
- Added class **UrbanSpaceEndUseValue**

DataTypes package:

- Added class **Suitability**

Core package:

- Class **ADE _CityObject**
 - Added property *suitability*
- Class **ADE _AbstractBuilding**
 - Renamed property *bdgFloorArea* to *bdgArea*
 - Renamed property *bdgHeightAboveGround* to *bdgHeight*
- Class **AbstractBuildingPartition**
 - Renamed property *floorArea* to *area*

Resources package:

- Added class **UrbanSpace**

2025-04-12**Core package:**

- Added class **AbstractLibrary**

Layered Construction package:

- Added class **LayeredConstructionLibrary**
- Added class **MaterialLibrary**

Schedules package:

- Added class **ScheduleLibrary**

CodeLists package:

- Class **FunctionInNetwork** renamed to **FunctionInNetworkValue**

Version 2.0 beta 6

2025-02-06**DataTypes package:**

- Dropped class **HeatExchangeType**
- Dropped class **OpticalProperties**

Core package:

- Class **ADE_CityObject**:
 - Renamed property from *operationSchedule* to *deviceOperation*
- Class **AbstractDevice**:
 - Changed type of property *heatDissipation* to gml:Measure
 - Added property *heatDissipationConvectiveFraction*
 - Added property *heatDissipationLatentFraction*
 - Added property *heatDissipationRadiantFraction*

Occupancy package:

- Class **UsageZone**:
 - Renamed property *averageInternalGains* to *internalGains* and changed type to gml:Measure
 - Added property *internalGainsConvectiveFraction*
 - Added property *internalGainsLatentFraction*
 - Added property *internalGainsRadiantFraction*
- Class **Occupants**:
 - Changed type of property *heatDissipation* to gml:Measure
 - Added property *heatDissipationConvectiveFraction*
 - Added property *heatDissipationLatentFraction*
 - Added property *heatDissipationRadiantFraction*

Layered Construction package:

- Class **LayeredConstruction**:
 - Dropped property *opticalProperties*
 - Added property *glazingRatio*
 - Added property *emissivity*
 - Added property *reflectance*
 - Added property *transmittance*
- Class **Layer**:
 - Changed multiplicity of property *thickness* to [1]
- Class **AbstractMaterial**:
 - Added property *libraryCode*

Version 2.0 beta 5

Resources package:

No changes since beta 4

Weather Station package:

No changes since beta 4

2025-01-30**Packages**

- Created TimeSeries package and moved all related classes into it
- Created Schedule package and moved all related classes into it
- Dropped Supporting classes package

CodeLists package:

- Class **CertificateTypeValue** renamed to **EnergyPerformanceCertificateTypeValue**

2025-01-28

CodeLists package:

- Added class **ScheduleTypeValue**

Enumerations package:

- Dropped class **DayTypeValue**

Supporting classes package:

- Dropped class **TimeSeriesSchedule**
- Dropped class **DailyPatternSchedule**
- Dropped class **PeriodOfYear**
- Dropped class **DailySchedule**
- Added class **AtomicSchedule**
- Added class **CompositeSchedule**
- Added class **ScheduleComponent**
- Class **AbstractSchedule**:
 - Added properties *type*, *startTime*, *startDay*, *startMonth*, *startYear*, *temporalExtent*

2025-01-22

Devices package:

- Class **AbstractSolarCollector**:
 - Added properties *azimuth* and *inclination*
 - Dropped property *surfaceGeometry*
 - Added properties *lod2MultiSurface* and *lod3MultiSurface*

2025-01-13

CodeLists package:

- Dropped class **ThermalBoundaryTypeValue**
- Added class **ShadingDeviceTypeValue**
- Added class **WeatherValueTypeValue**

DataTypes package:

- Moved class **PeriodOfYear** from package DataTypes to Supporting classes
- Moved class **DailySchedule** from package DataTypes to Supporting classes
- Class **AbstractQualifiedAttribute**:
 - Added properties *source* and *value*
- Classes **QualifiedHeight**, **QualifiedArea**, **QualifiedVolume**:
 - Dropped property *value*

Supporting classes package:

- Class **AbstractTimeSeries**:
 - Moved back from package Core to Supporting classes
 - Changed multiplicity of property *acquisitionMethod* to [1]
 - Changed multiplicity of property *interpolationType* to [1]
- Class **TypicalValuesTimeSeries**:
 - Added properties *startTime*, *startDay*, *startMonth*
- Class **TypicalValuesTimeSeriesFile**:
 - Added properties *startTime*, *startDay*, *startMonth*
- Added class **DualValueSchedule**:

- Class **PeriodOfYear**:
 - Changed stereotype from *dataType* to *featureType*
 - Added properties *startDay*, *startMonth*, *endDay*, *endMonth*
 - Dropped property *period*
- Class **DailySchedule**:
 - Changed stereotype from *dataType* to *featureType*
 - Added property *repetition*
 - Renamed property from *schedule* to *timeSeries*

Layered Construction package:

- Added class **ReverseLayeredConstruction**
- Dropped class **LayerComponent**
- Class **Layer**:
 - Removed property *layerComponent*
 - Added property *thickness*
 - Added property *material*
- Class **SolidMaterial**:
 - Renamed property from *specificHeat* to *specificHeatCapacity*
 - (Re)added and renamed property from *conductivity* to *thermalConductivity*
 - (Re)added property *embodiedEnergy*
 - (Re)added property *embodiedCarbon*
 - (Re)added property *porosity*

Core package:

- Class **AbstractDevice**:
 - Class is renamed from **AbstractEnergySystem** to **AbstractDevice**
 - Added property *heatDissipation*
 - Added property *operationSchedule*
 - Added property *efficiencyIndicator*
 - Added property *installedPower*
 - Added property *nominalEfficiency*
- Class **AbstractTimeSeries**:
 - Moved back from package Core to Supporting classes
- Class (ADE) **_CityObject**:
 - Renamed property from *energySystem* to *device*
- Class **OperationSchedule**:
 - Moved from package Devices to Core
- Class **WeatherData**:
 - Renamed property from *timeseries* to *timeDependentValues*
 - Renamed property from *singleValue* to *yearlyValue*
 - Renamed property from *singleValueType* to *valueType*
 - Changed type of property *valueType* from *CharacterString* to *CodeList*
- Added class **AbstractBuildingPartition**
- Classes **AbstractBuildingUnit**, **AbstractUsageZone**, **AbstractThermalZone**
 - Changed parent class

Building physics package:

- Dropped class **ThermalBoundary**
- Dropped class **ThermalOpening**
- Class **ThermalZone**
 - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volumeGeometry* (now inherited via **AbstractBuildingPartition**)
 - Removed property *boundedBy*
 - Added property *heatCapacity*
 - Added property *coincidesWithLod2Hull*
 - Added property *coincidesWithLod3Hull*
 - Added property *thermalBoundary*
- Added class **PartyWallSurface**

- Added class **BasementCeilingSurface**
- Added class **IntermediateFloorSurface**
- Added class **AtticFloorSurface**
- Class (ADE) **BoundarySurface**:
 - Renamed property from *bdgBdrySrfWindowToWallRatio* to *bdgBdrySrfOpeningToSurfaceRatio*
 - Renamed property from *bdgBdrySrfArea* to *bdgBdryTotalSurfaceArea*
 - Added property *bdgBdryOpaqueSurfaceArea*
 - Added property *bdgBdrySrfThickness*
 - Added property *bdgBdrySrfHeatCapacity*

Occupancy package:

- Dropped class **AbstractFacilities**
- Dropped class **DHWFacilities**
- Dropped class **LightingFacilities**
- Dropped class **ElectricalAppliances**
- Dropped class **GenericFacilities**
- Class **BuildingUnit**:
 - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
 - Removed property *equippedWith*
- Class **UsageZone**:
 - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volumeGeometry* (now inherited via **AbstractBuildingPartition**)
 - Removed property *equippedWith*

Devices package:

The former EnergySystems package is renamed to Devices package

- Class **OperationSchedule**:
 - Moved from package Devices to Core
- Dropped class **AbstractEnergyConversionSystem**
 - Now all subclasses are derived directly from **AbstractDevice**
- Dropped class **AbstractStorageSystem**
 - Now all subclasses are derived directly from **AbstractDevice**
- Class **GenericDevice**:
 - Renamed from class **GenericEnergyConversionSystem**
- Class **ElectricalStorageDevice**:
 - Renamed from class **PowerStorageSystem**
- Class **ThermalStorageDevice**:
 - Renamed from class **ThermalStorageSystem**
- Class **AbstractSolarCollector**:
 - Renamed from class **AbstractSolarEnergySystem**
- Class **PhotovoltaicThermalCollector**:
 - Renamed from class **PhotovoltaicThermalSystem**
- Class **PhotovoltaicCollector**:
 - Renamed from class **PhotovoltaicSystem**
- Class **SolarThermalCollector**:
 - Renamed from class **SolarThermalSystem**
- Class **GenericSolarCollector**:
 - Renamed from class **GenericSolarEnergySystem**
- Added class **LightingDevice**
- Added class **GenericElectricalDevice**
- Added class **MovableShadingDevice**

Urban Function Areas package:

- Class **UrbanFunctionArea**:
 - Removed property *rank*
-

Version 2.0 beta 4

2024-09-25

- Codelist **ThermalBoundaryTypeValue**
 - Renamed *outerWall* to *exteriorWall*
 - Renamed *intermediaryFloor* to *intermediateFloor*

2024-09-20

- Minor edits to the layout of the UML diagrams
- Polished Codelist **ThermalBoundaryTypeValue**

2024-08-12

- Class **ADE CityObject**:
 - Renamed association *connectedTo* to *utilityNetworkConnection*
 - Remodelled association class **CityObjectRelation**
- Class **AbstractTimeSeries**:
 - Moved to Core module
- Class **WeatherData**:
 - renamed property *valuesList* to *timeSeries*
 - made the association to **AbstractTimeSeries** explicit
- Added some values to codelist **RelationTypeValue**
- Class **TimeSeriesSchedule**:
 - Changed association to **AbstractTimeSeries** to Composition
- Class **DailySchedule**:
 - Changed association to **AbstractTimeSeries** to Composition
- Class **SensorConnection**:
 - Changed type of *uom* to *UnitOfMeasure*
 - Renamed *linkToSensorDescription* to *linkToSensorDescription*
- Class **AbstractSolarEnergySystem**:
 - Removed association *installedOn* (now taken over by **CityObjectRelation**)
- Class **Energy**:
 - Changed multiplicity of *EnergyCarrier* to [0..1]
- Class **Waste**:
 - Changed multiplicity of *isDanerous* and *isRecyclable* to [0..1]

This is the last page of this document. It is left intentionally blank